

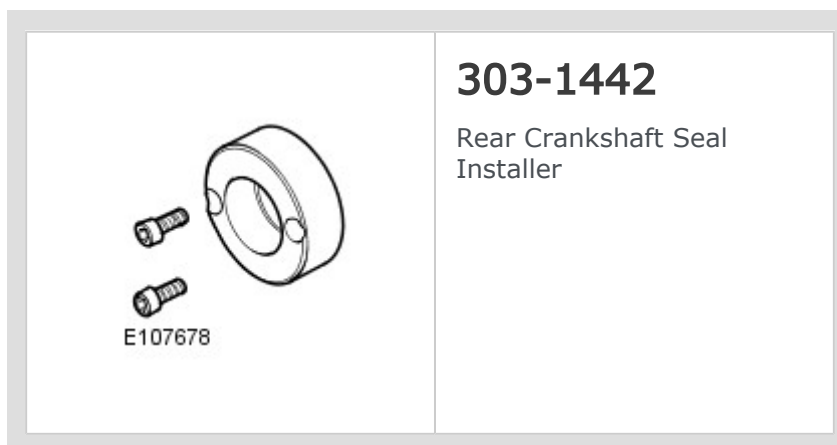
PUBLISHED: 31-AUG-2017
2014.0 RANGE ROVER (LG), 303-01

ENGINE - V8 S/C 5.0L PETROL

OIL PAN EXTENSION (G1509353)

REMOVAL AND INSTALLATION

SPECIAL TOOL(S)



PART(S)

STEP	PART NAME	QUANTITY
Installation Step 24	Fuel rail supply pipe(s)	1
Installation Step 33	Flexplate bolts	1
Installation Step 40	Fuel injector high pressure pipe(s)	1

REMOVAL

NOTES:

- Some components shown removed for clarity.
- Some variation in the illustrations may occur, but the essential information is always correct.

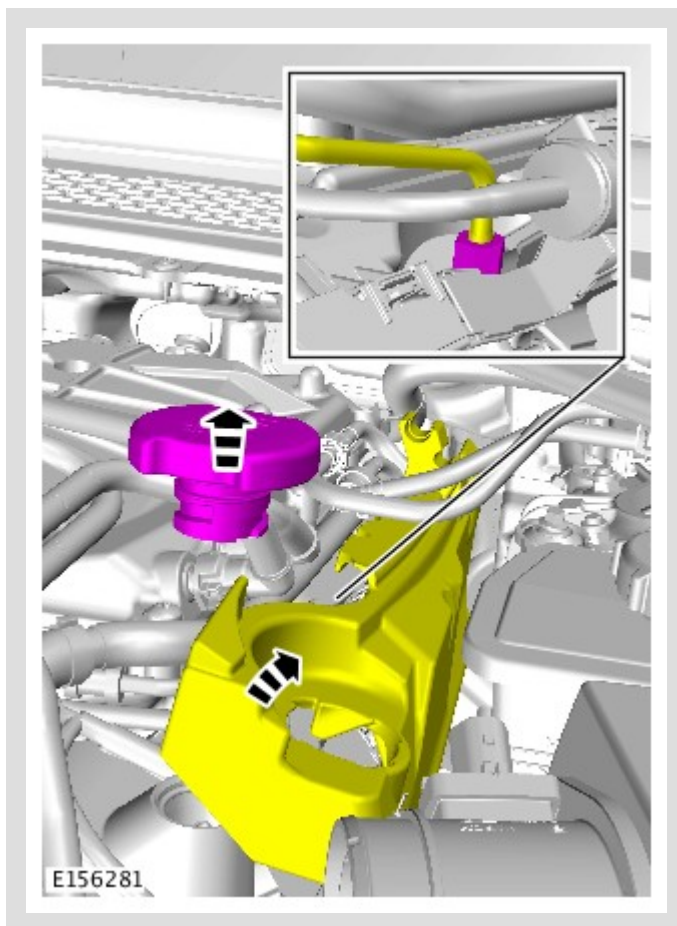
1. Disconnect the battery ground cable.
Refer to: Specifications (414-01 Battery, Mounting and Cables, Specifications).

2. Refer to: Engine Oil Draining and Filling (303-01 Engine - V8 S/C 5.0L Petrol, General Procedures).

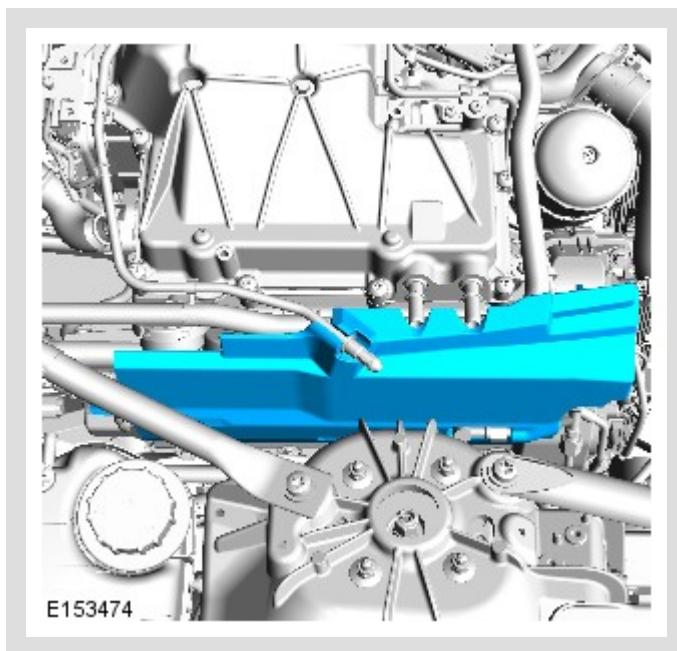
3. Refer to: High-Pressure Fuel System Bleeding (310-00 Fuel System - General Information, General Procedures).

4. Refer to: Fuel Injection Component Cleaning (303-04 Fuel Charging and Controls - V6 S/C 3.0L Petrol, General Procedures).

5.  **CAUTION:**
Be prepared to collect escaping fuel.



Loosen the high-pressure fuel supply line.



7.



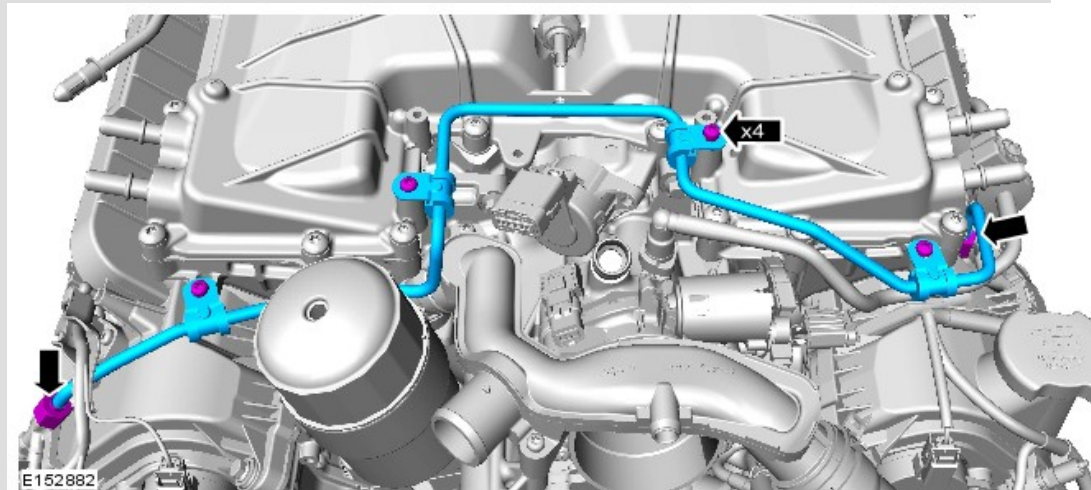
WARNING:

Do not smoke or carry lighted tobacco or open flame of any type when working on or near any fuel related components. Highly flammable mixtures are always present and may ignite. Failure to follow these instructions may result in personal injury.

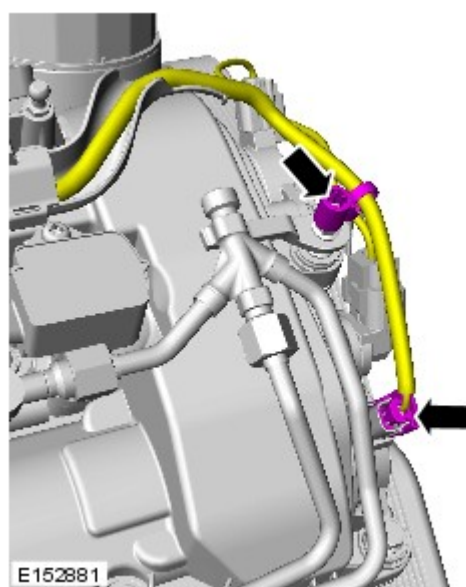


CAUTIONS:

- Be prepared to collect escaping fuel.
- Remove and discard the high-pressure fuel supply lines.



8.



9.



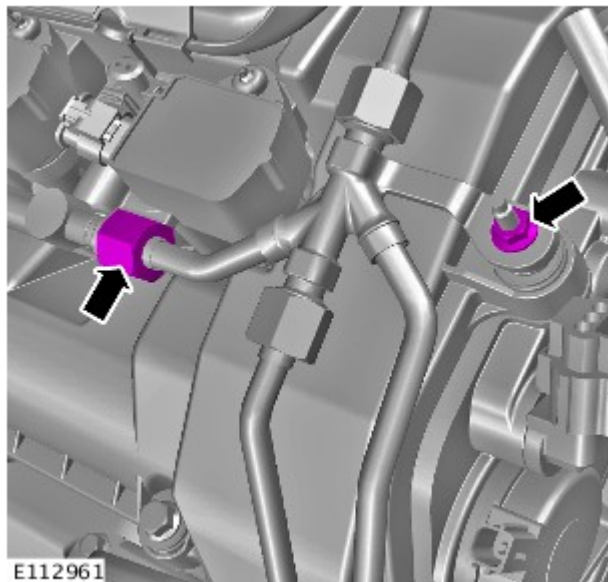
WARNING:

Do not smoke or carry lighted tobacco or open flame of any type when working on or near any fuel related components. Highly flammable

mixtures are always present and may ignite. Failure to follow these instructions may result in personal injury.

⚠ CAUTIONS:

- Be prepared to collect escaping fuel.
- Before disconnecting any components, make sure the area is clean and free from foreign material. When disconnected all openings must be sealed.



10.

⚠ WARNING:

Do not work on or under a vehicle supported only by a jack. Always support the vehicle on safety stands.

Raise and support the vehicle.

11.

Refer to: Generator (414-02 Generator and Regulator - V8 S/C 5.0L Petrol, Removal and Installation).

12.

Refer to: Differential Case (205-03 Front Drive Axle/Differential, Removal and Installation).

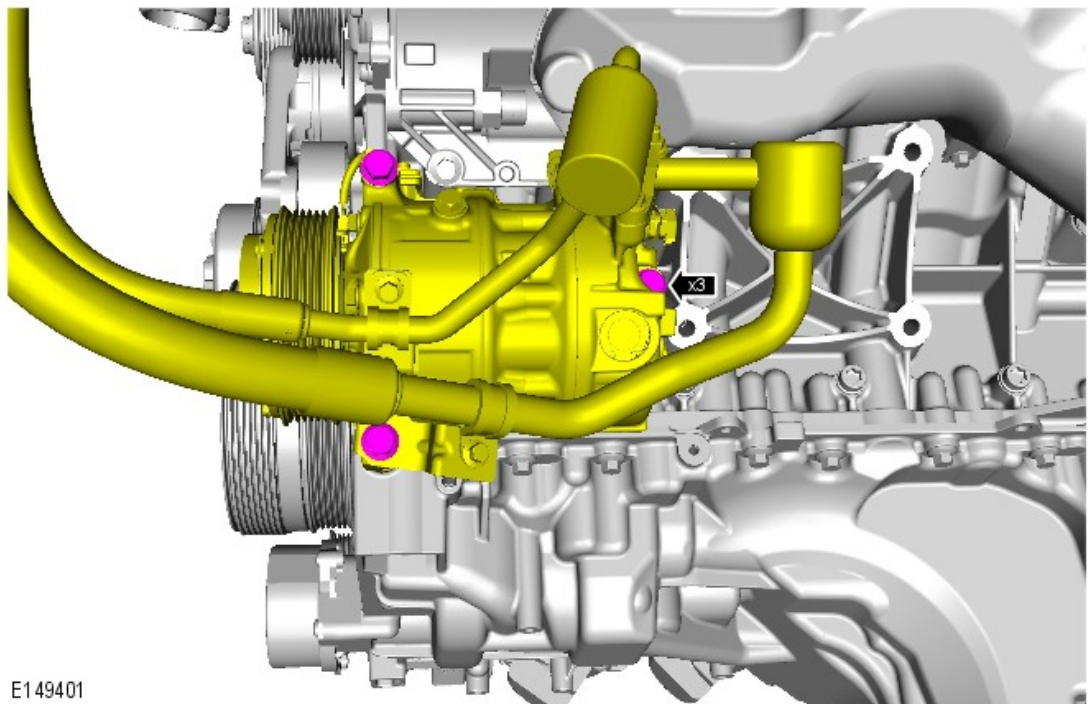
13. Refer to: Transmission - V6 S/C 3.0L Petrol /V8 S/C 5.0L Petrol, Vehicles With: Twin Speed Transfer Case (307-01 Automatic Transmission/Transaxle - Vehicles With: 8HP70 8-Speed Automatic Transmission AWD, Removal).

14. Refer to: Starter Motor (303-06 Starting System - V8 S/C 5.0L Petrol, Removal and Installation).

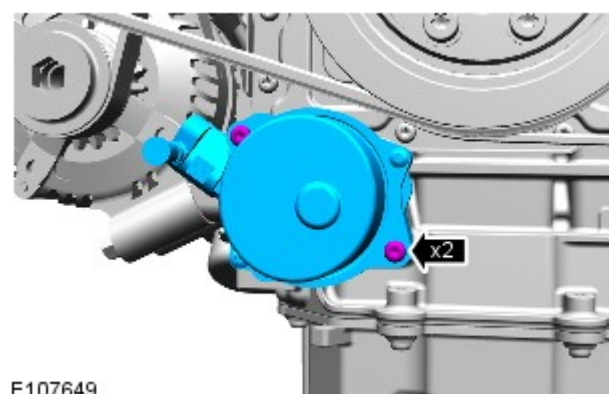
15.

 NOTES:

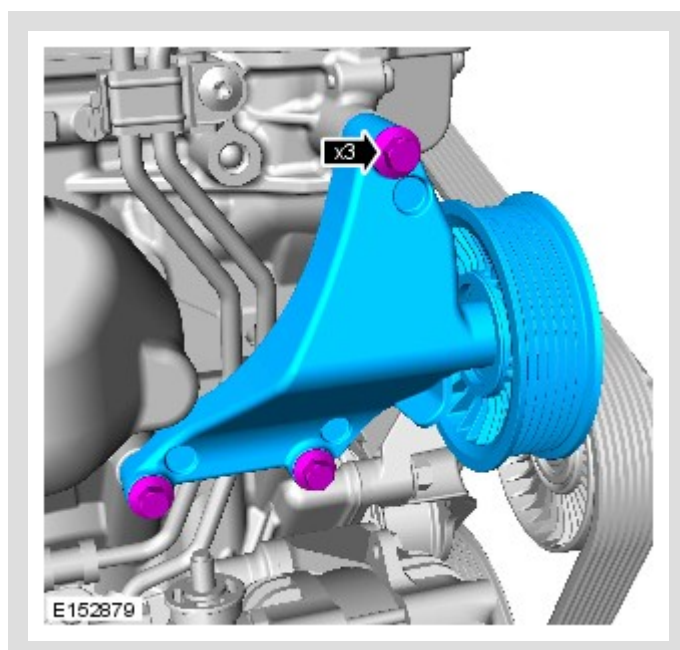
- Using suitable cable tie secure the A/C compressor.
- Engine shown removed for clarity.



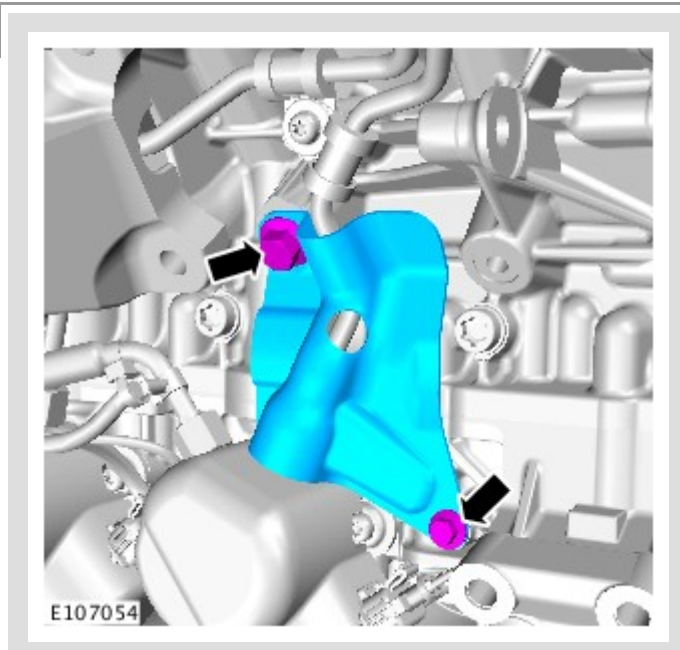
16.



17.



18.



19.

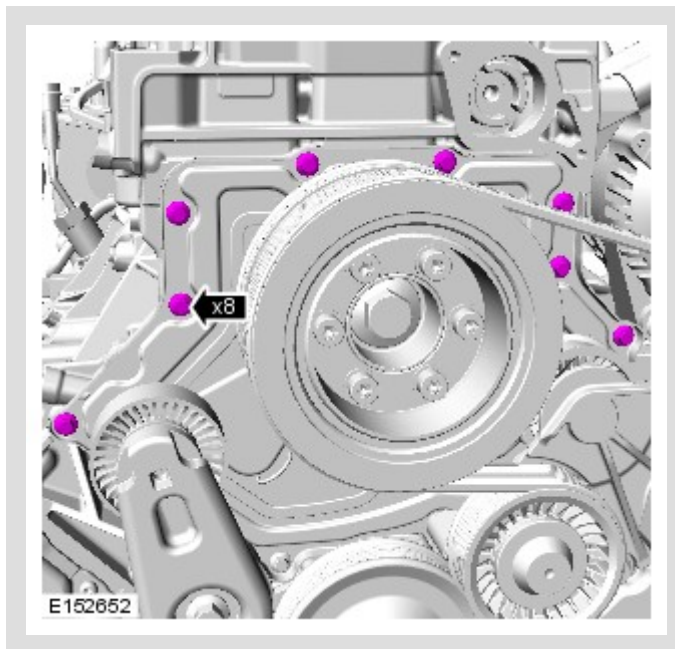


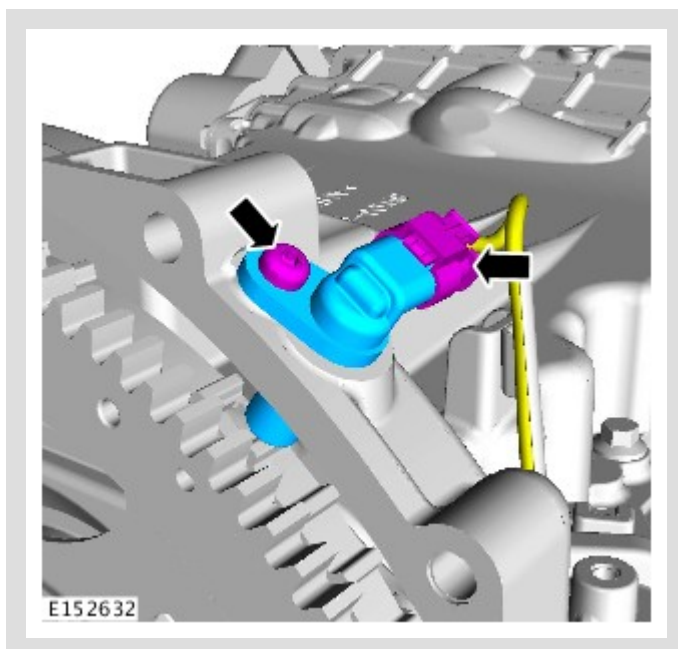
CAUTION:

Only remove the bolts shown in the illustration, do not remove the crankshaft damper.

 NOTE:

Engine shown removed for clarity.

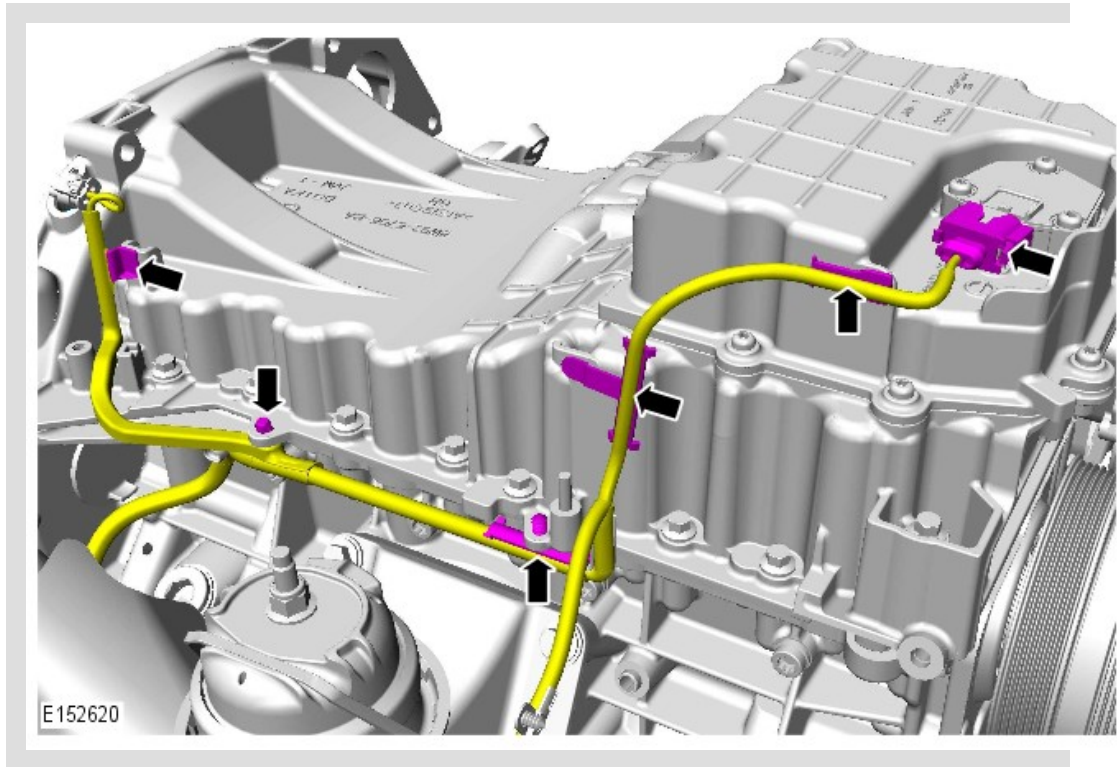




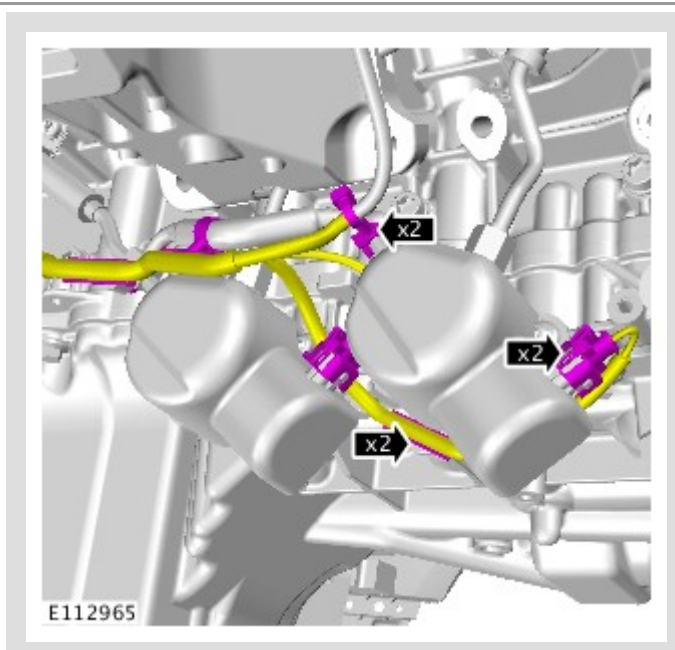
21.

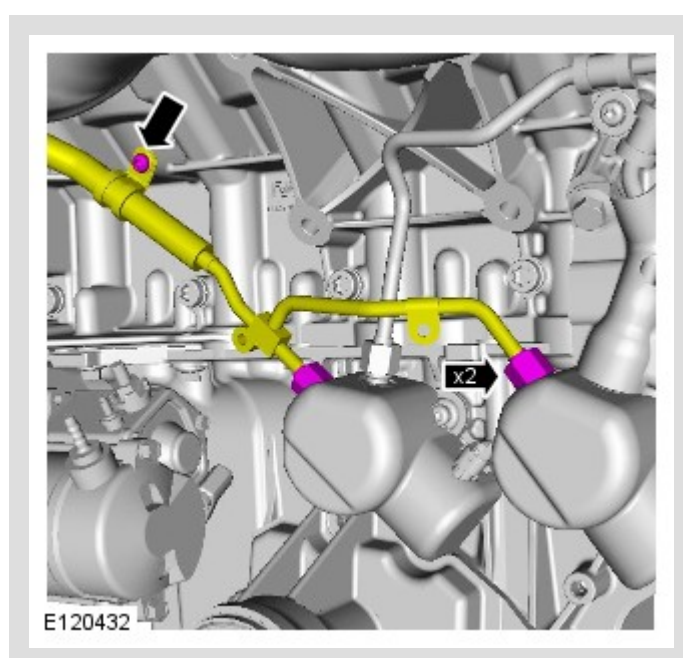
 NOTE:

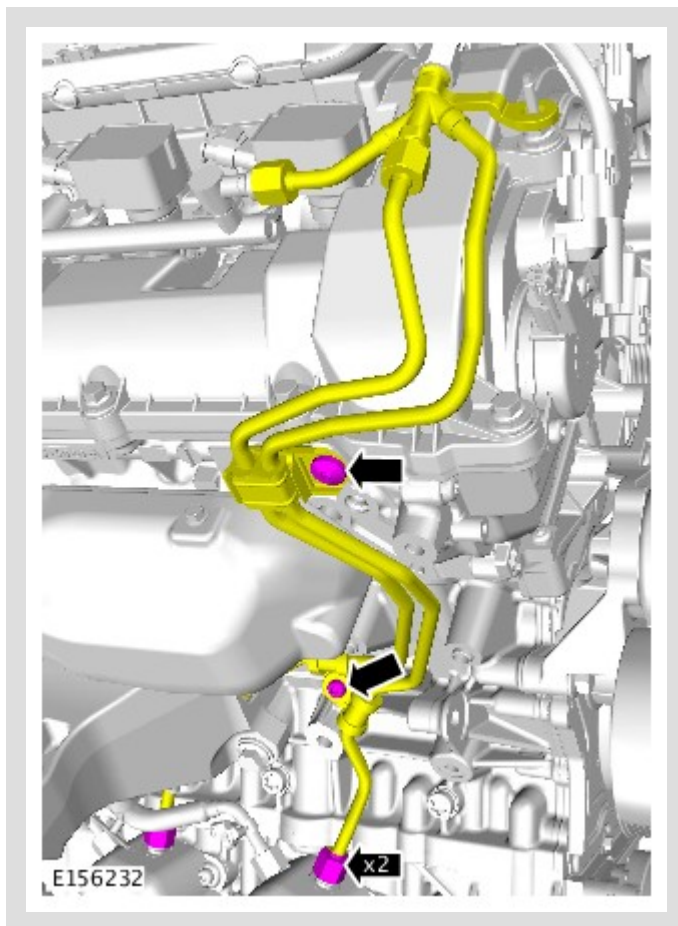
Engine shown removed for clarity.

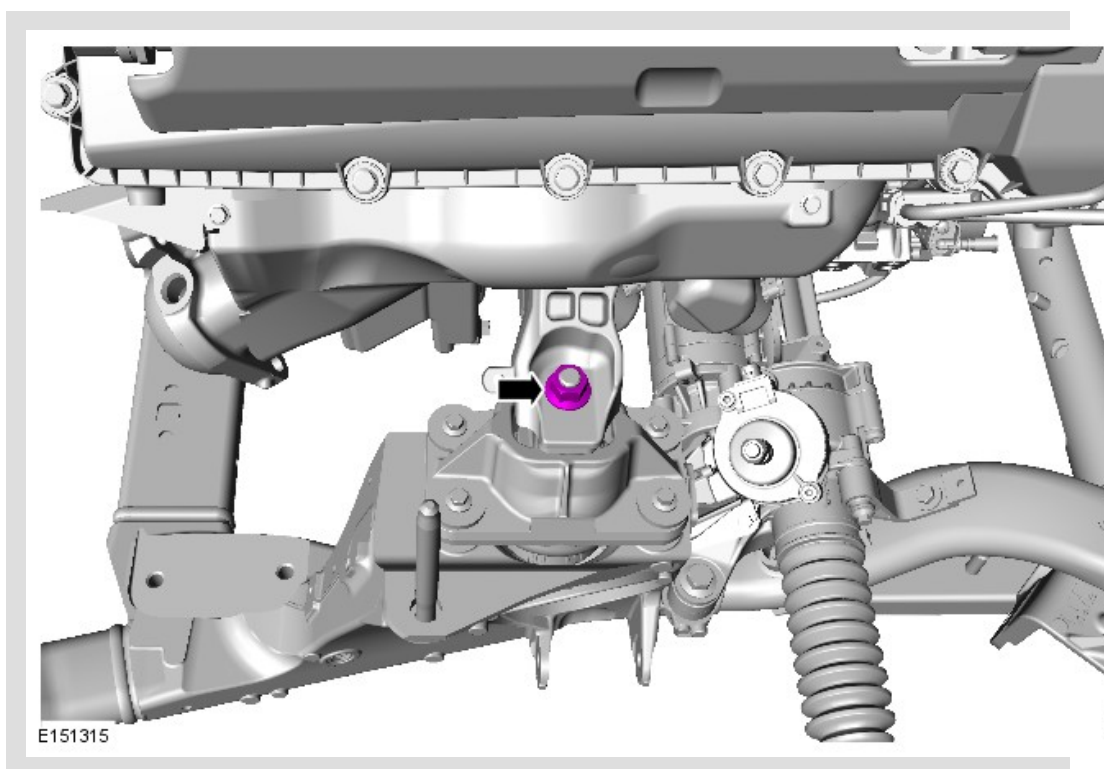


22.





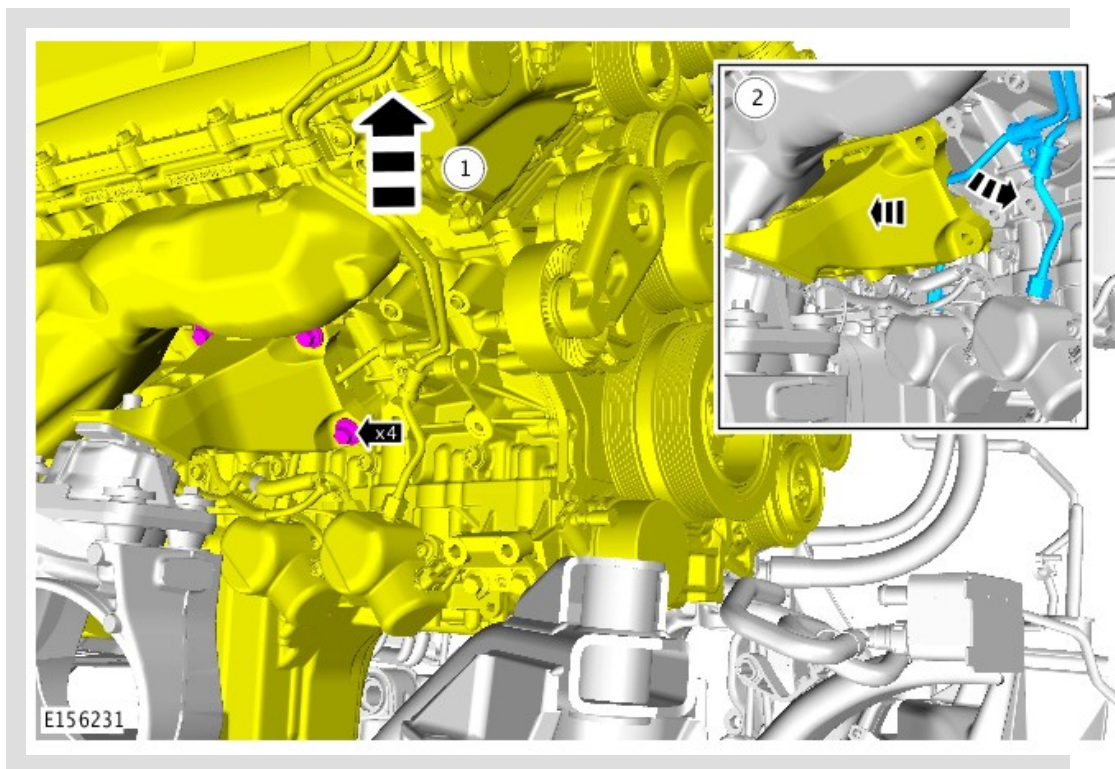




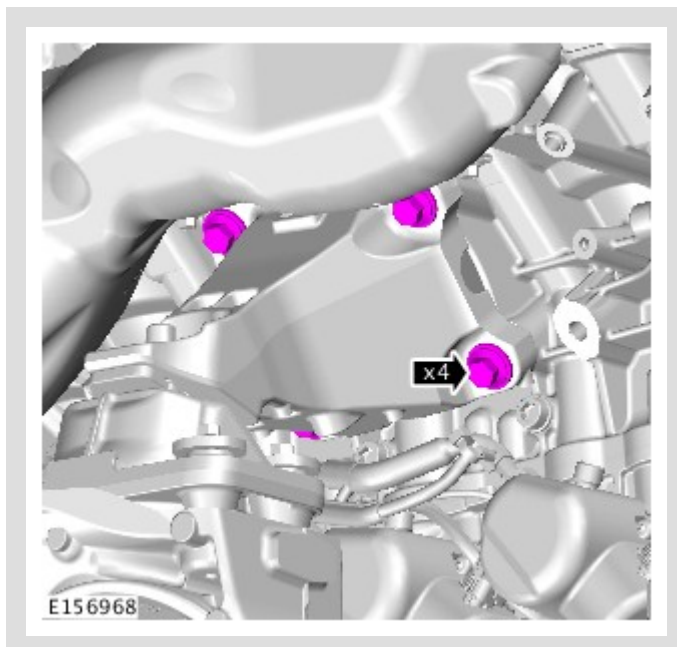
26.

ⓘ CAUTION:

Use a wooden block to protect the oil pan when supporting the engine.

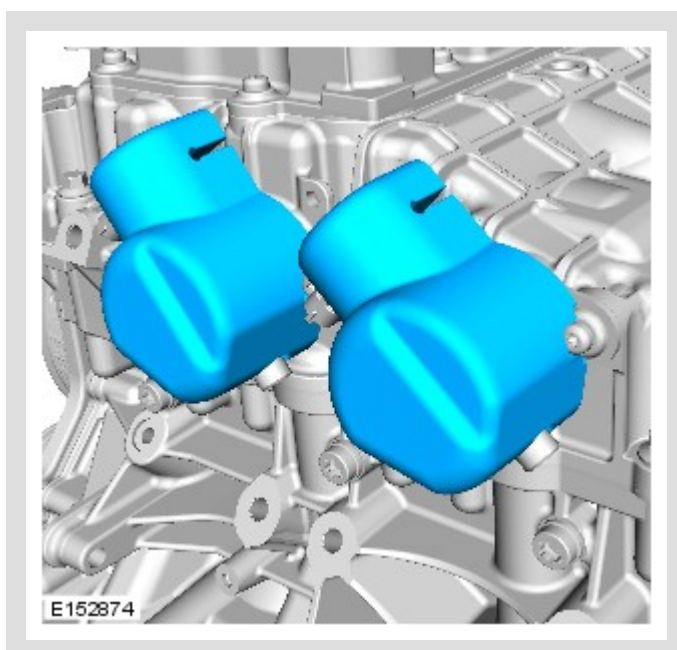


Using a suitable hydraulic jack, raise and support the engine to release the fuel pipe.



Carefully lower the engine onto the engine mounts.
Torque: 20 Nm

28.

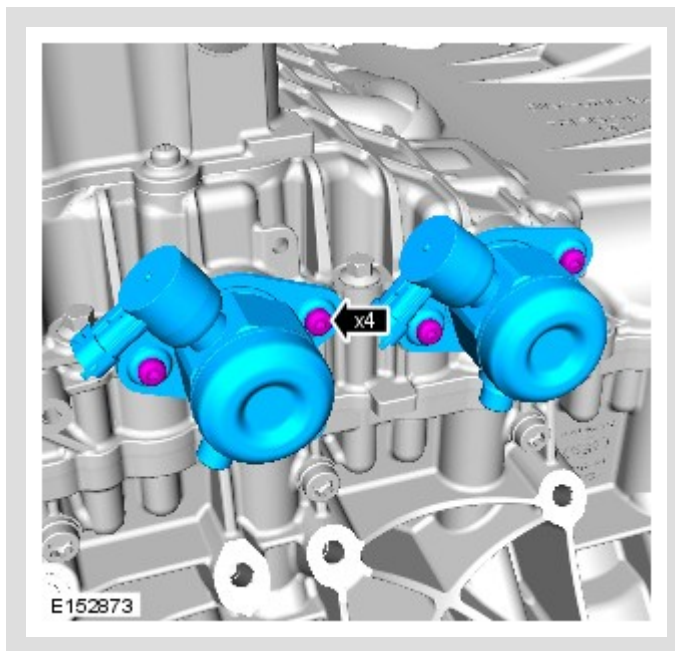


29.



CAUTION:

Be prepared to collect escaping fluids.



30.



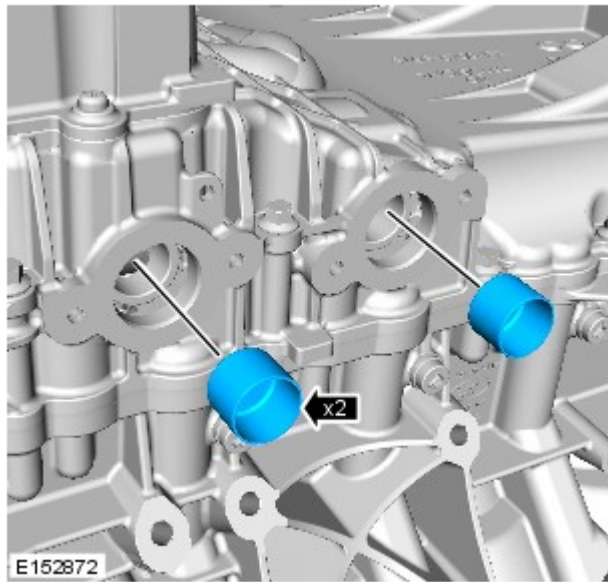
CAUTION:

Be prepared to collect escaping fluids.



NOTE:

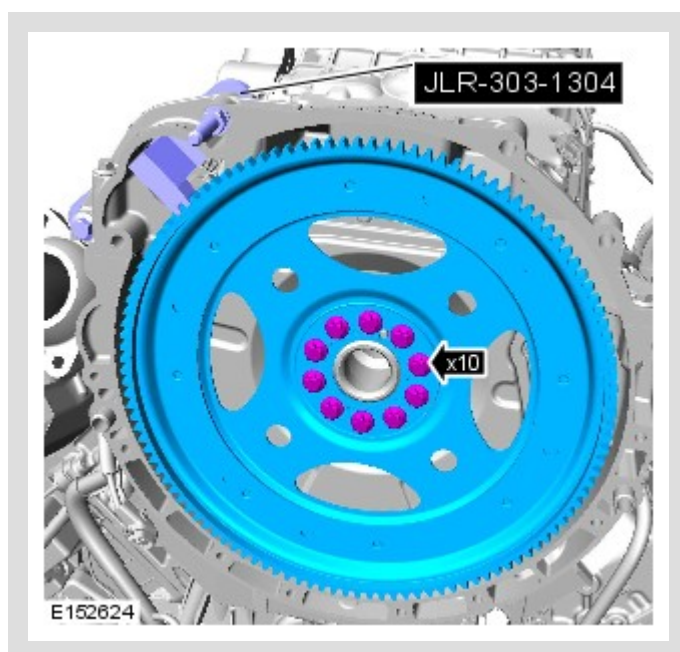
Note the orientation of the component prior to removal.



31.

⚠ CAUTION:

The bolts can only be used 3 times, mark the bolts with a center punch. If 2 punch marks are visible, discard the bolts.



Install the special tool as shown.

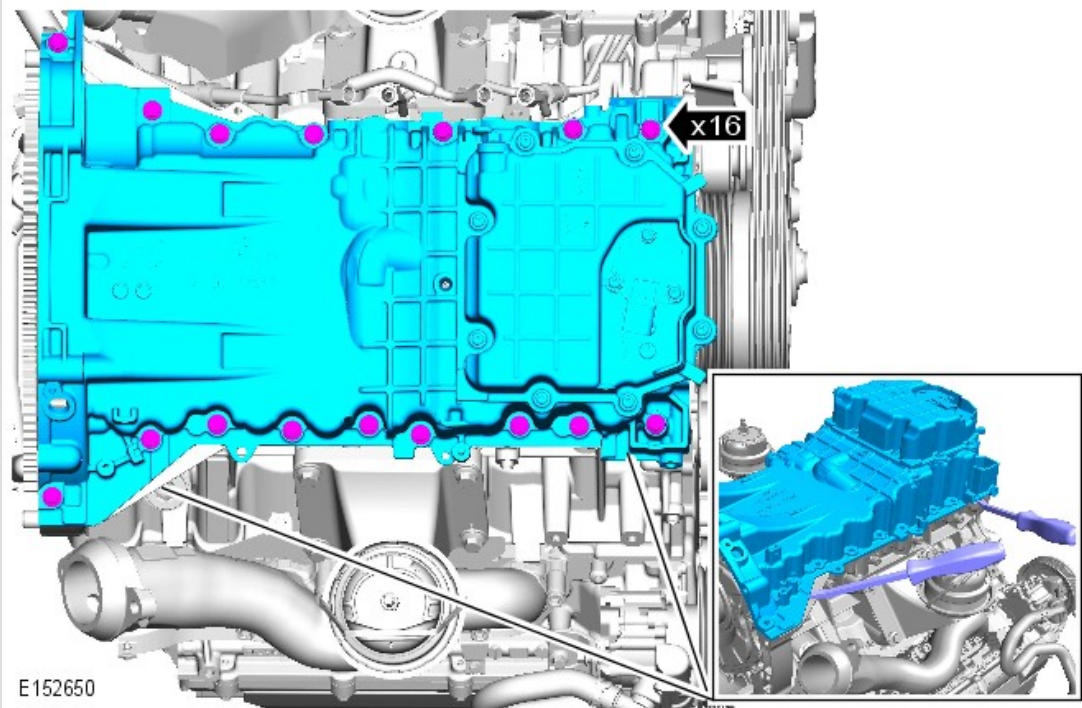
32.



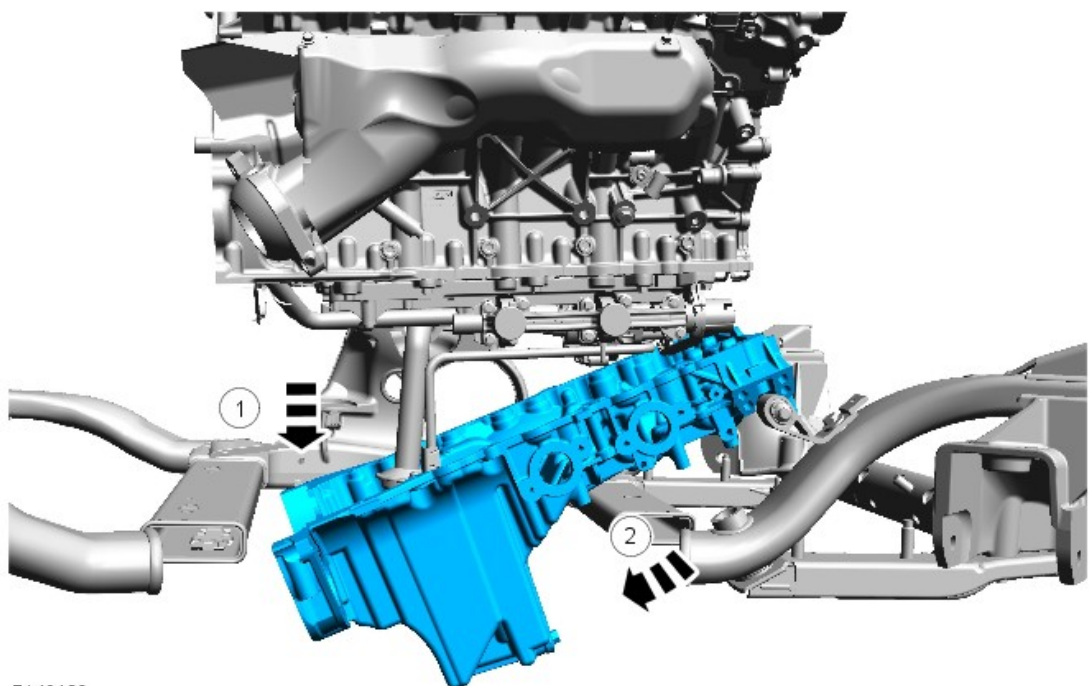
33.

△ NOTE:

Make sure to use the aluminium lug provided on the oil pan extension to lever against.



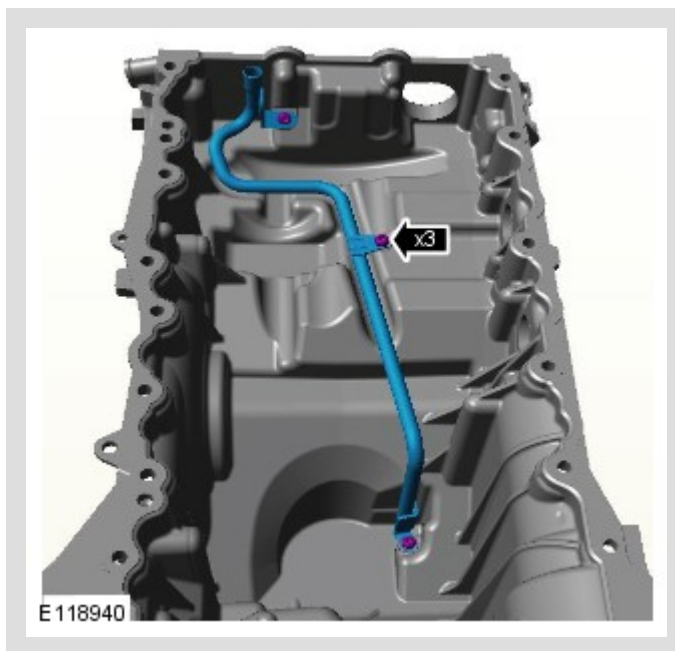
34.



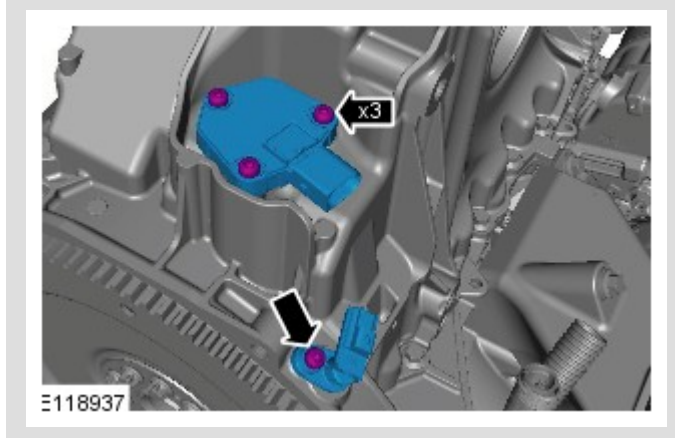
35.

 NOTE:

Do not disassemble further if the component is removed for access only.

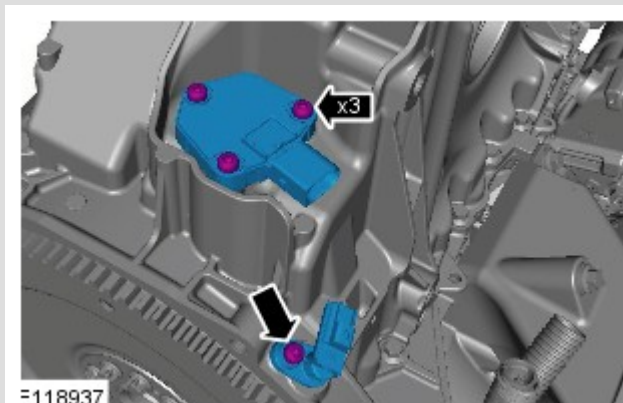


36.



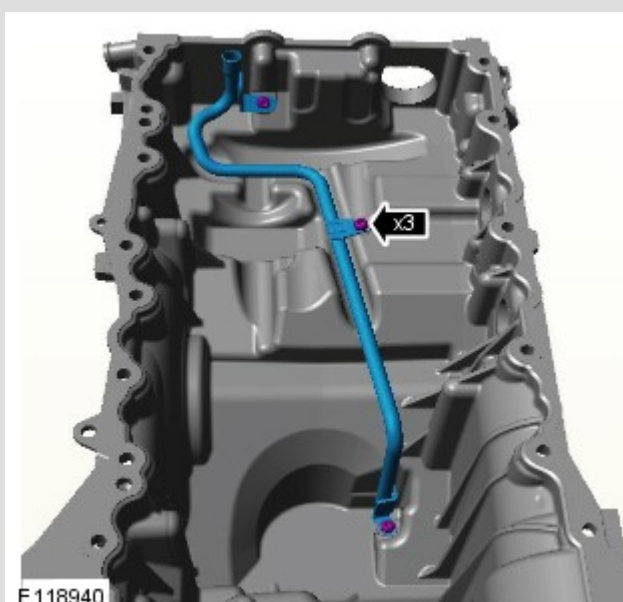
INSTALLATION

1.



■ Torque: 11 Nm

2.

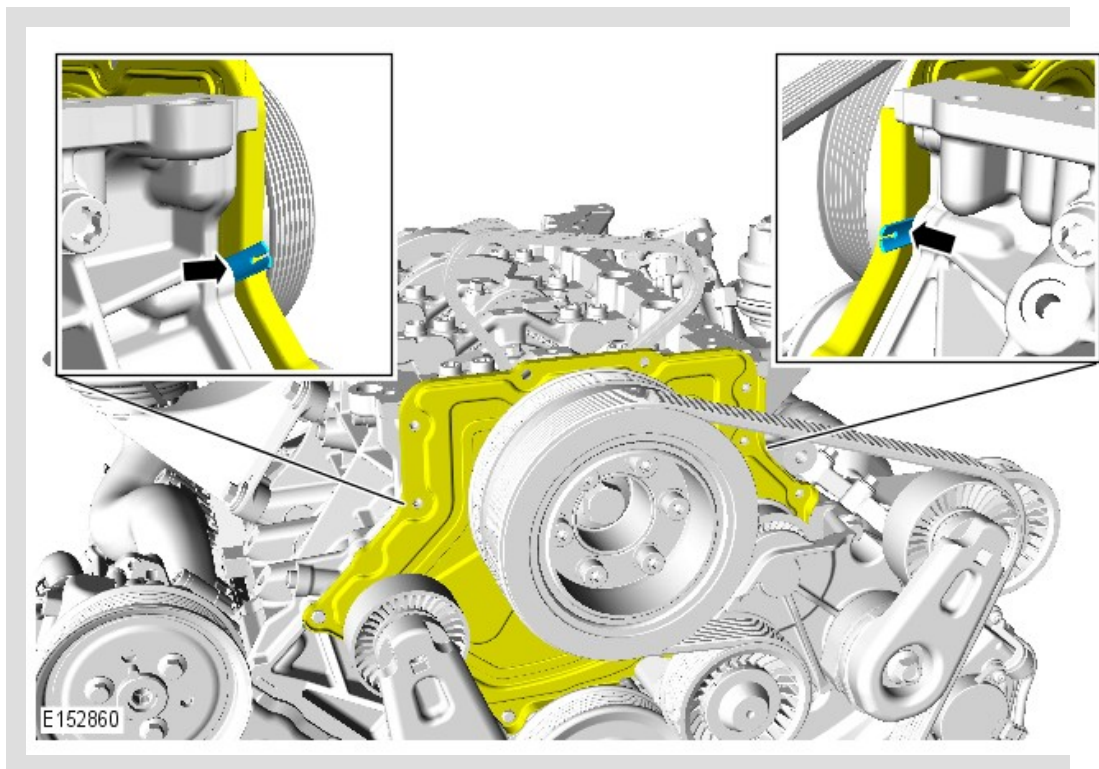


Torque: 10 Nm

3.

△ NOTE:

Engine shown removed for clarity.



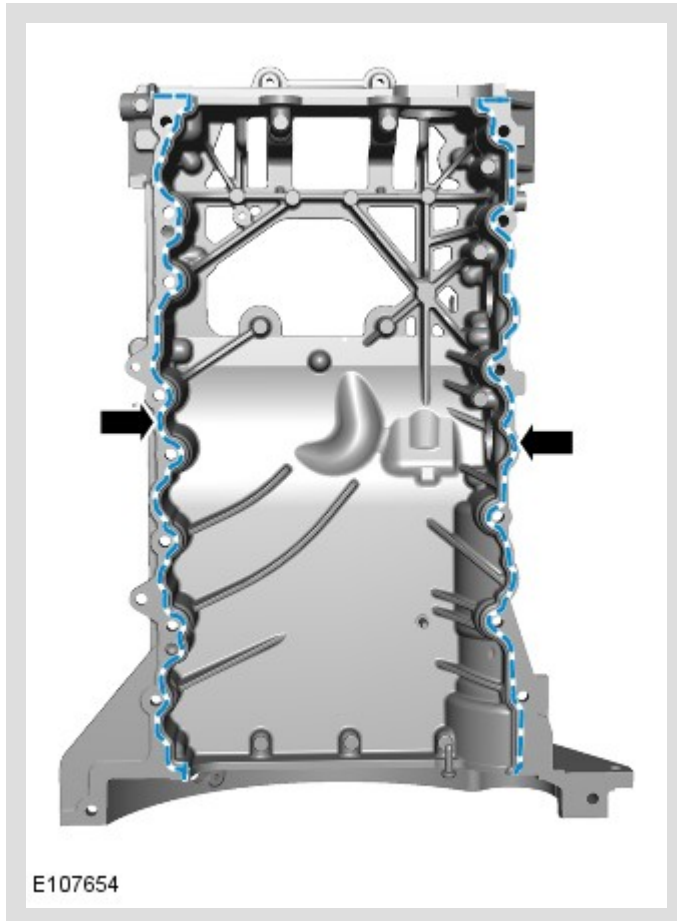
Using a folded cable tie, pack the two areas between the front cover plate and the cylinder block as illustrated. Care must be taken to avoid bending the front cover.

4.

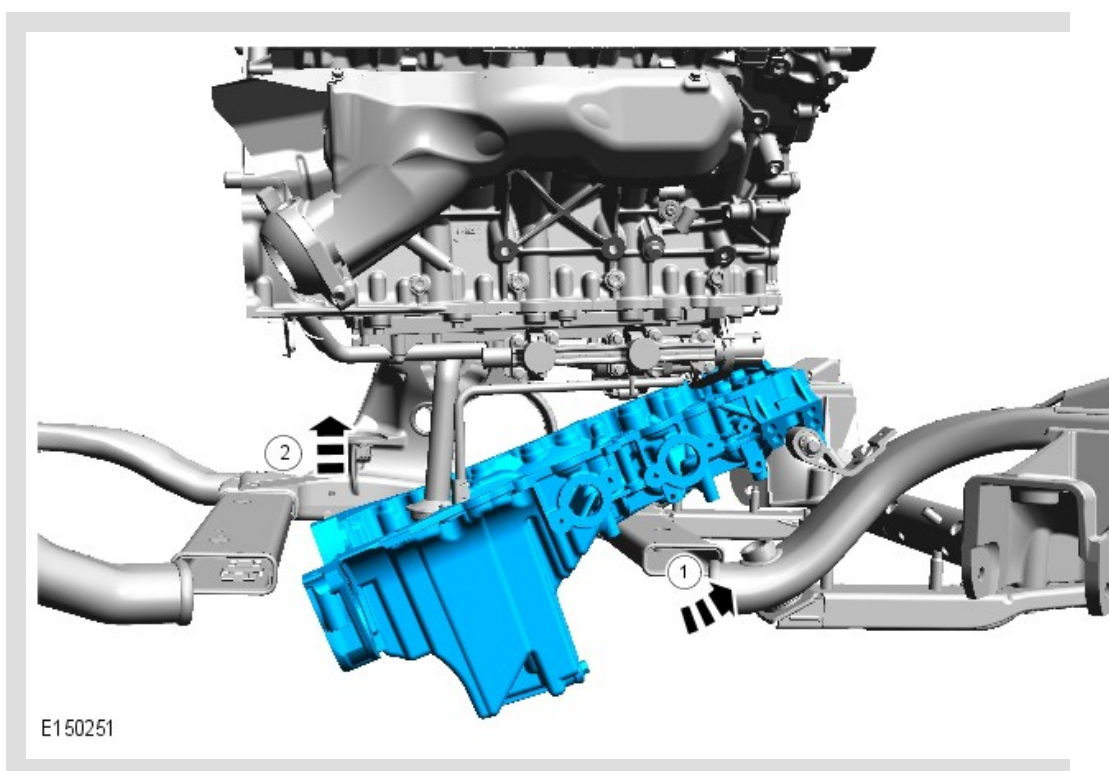


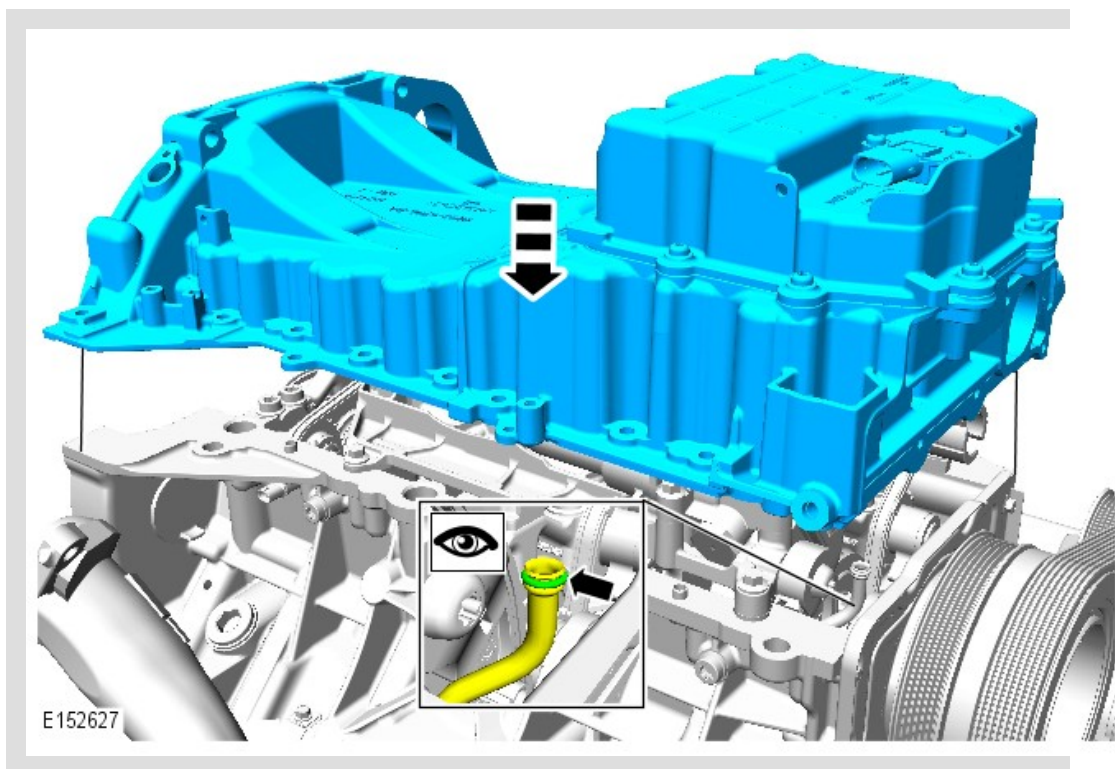
CAUTIONS:

- Make sure that the mating faces are clean and free of foreign material.
- Use only a plastic scraper when removing the sealing material.



- Apply RTV sealant WSE-M4G323-A6 (Loctite 5901G) to the areas shown, and tighten the bolts within 7 minutes.

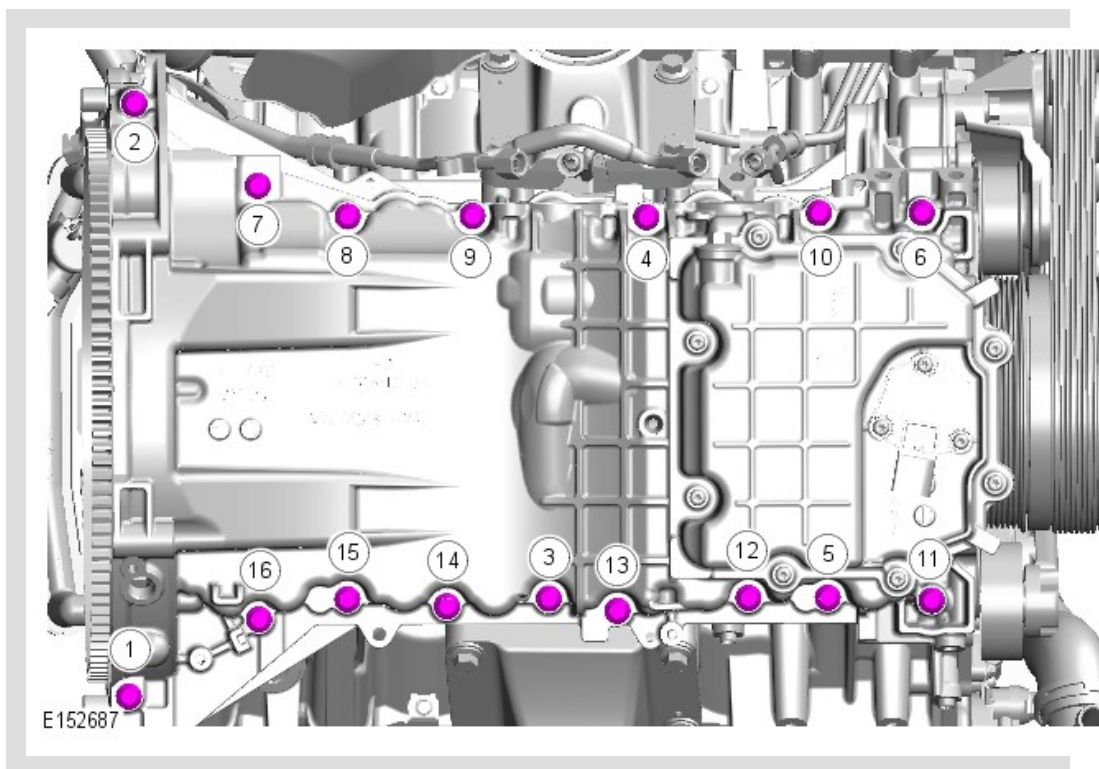




7.

 NOTE:

Tighten the bolts in the indicated sequence.



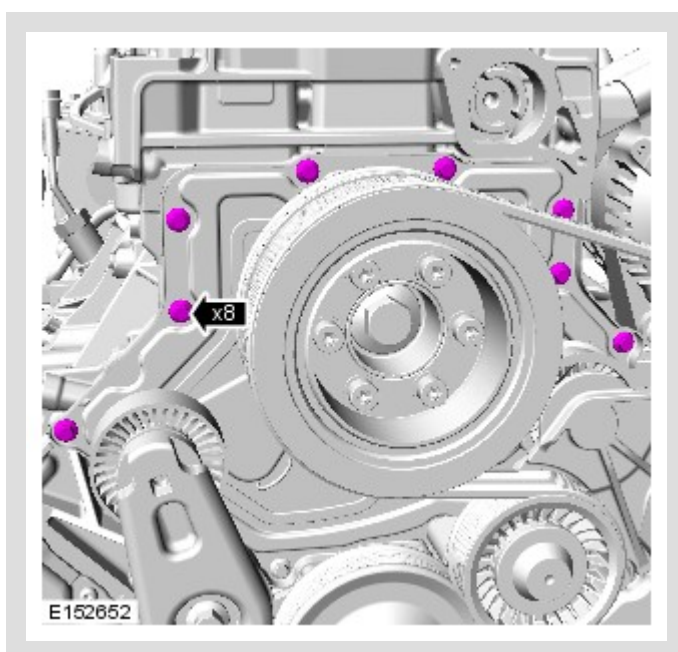
Torque: 25 Nm

8.

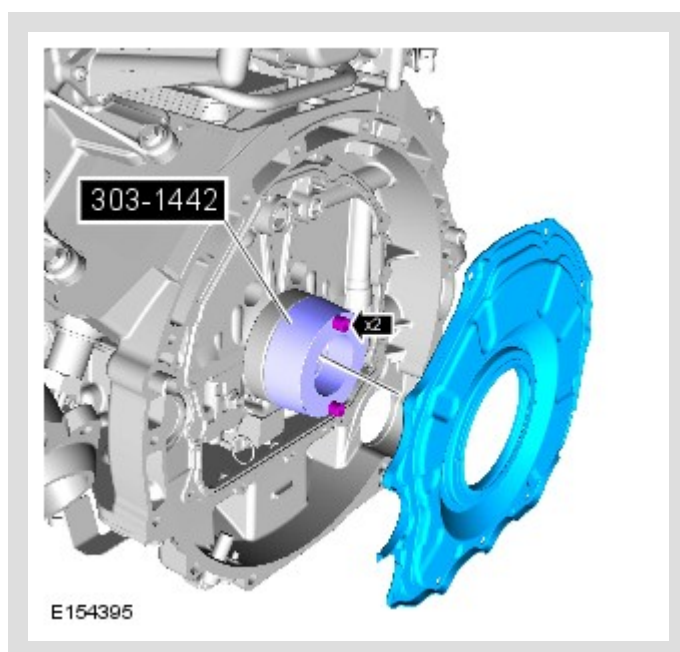


CAUTION:

Make sure the front cover spacers are removed before installing the bolts.



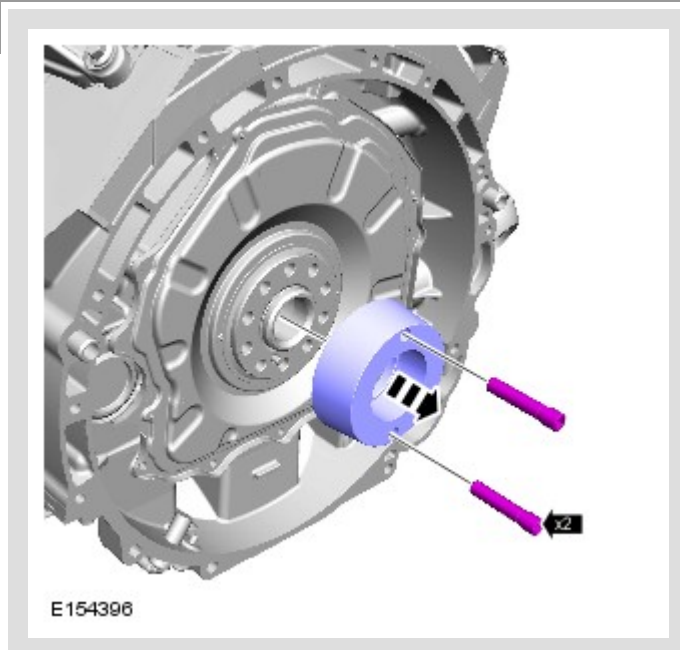
Torque: 12 Nm



Install the special tool.

Special Tool(s): [303-1442](#)

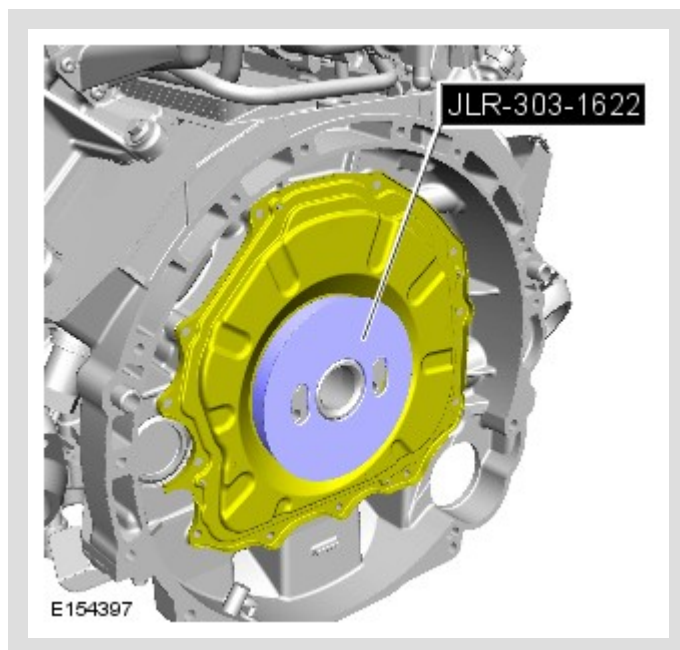
10.



Remove the special tool.

Special Tool(s): [303-1442](#)

11.

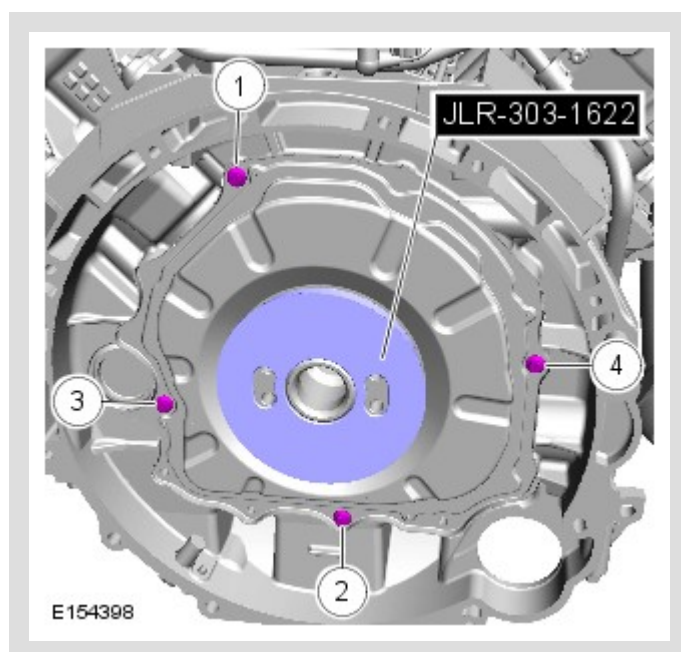


Align the engine rear oil seal plate using the special tool.

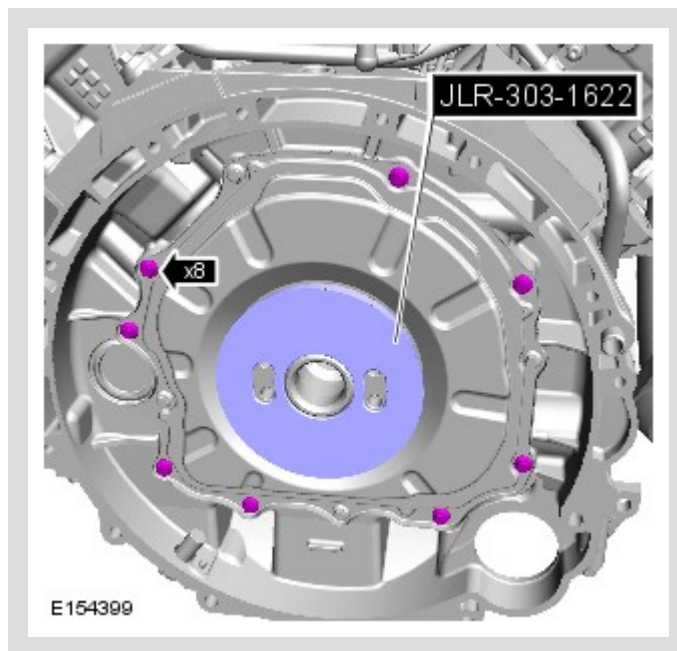
12.

 NOTE:

Tighten the bolts in the indicated sequence.



Torque: 11 Nm

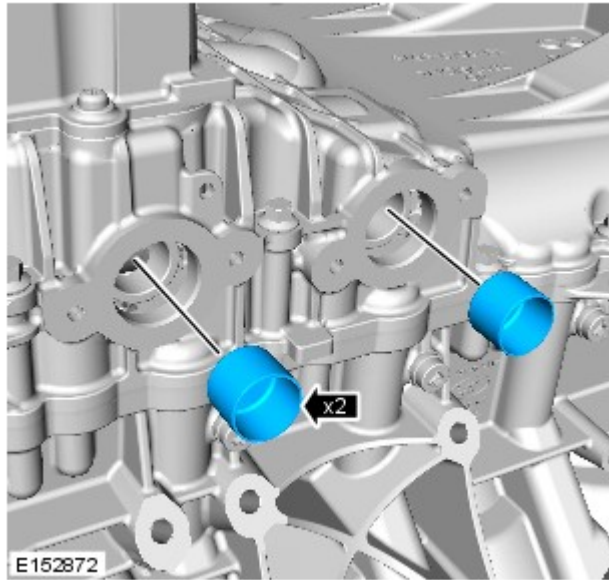


Torque: 11 Nm

-
14. Remove the special tools.

-
15.  **NOTE:**

Lubricate the fuel rail high-pressure fuel pump bucket with clean engine oil.



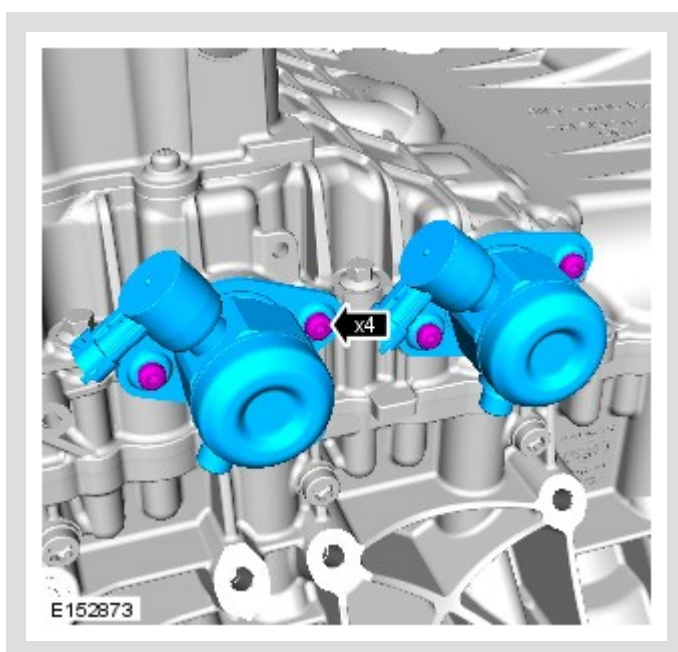
16.

⚠ CAUTION:

Tighten the Torx screws a turn at a time until the correct torque is achieved.

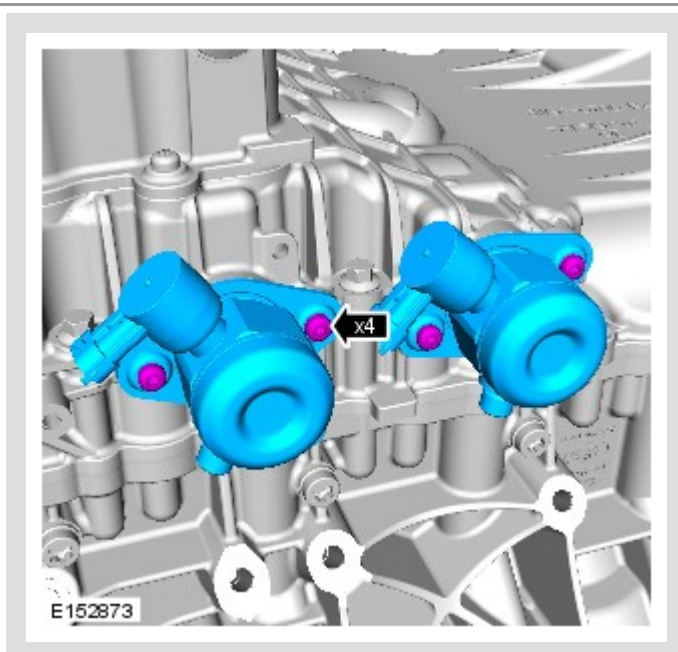
⚠ NOTE:

Lubricate the fuel rail high-pressure fuel pump O-ring seal with clean engine oil.



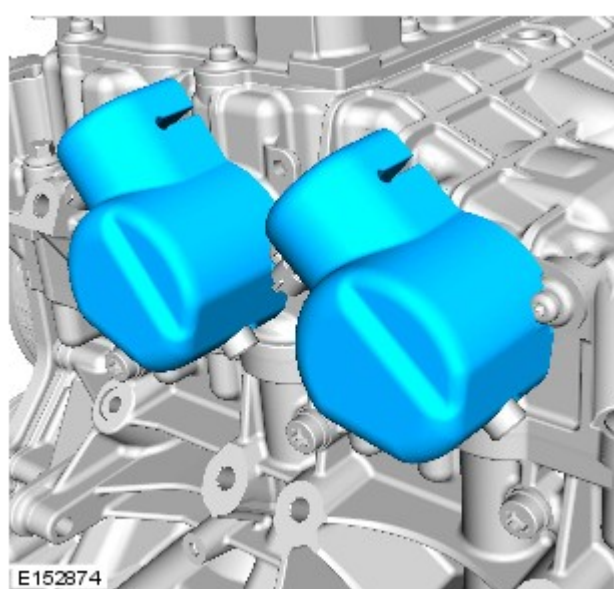
Torque: 12 Nm

17.

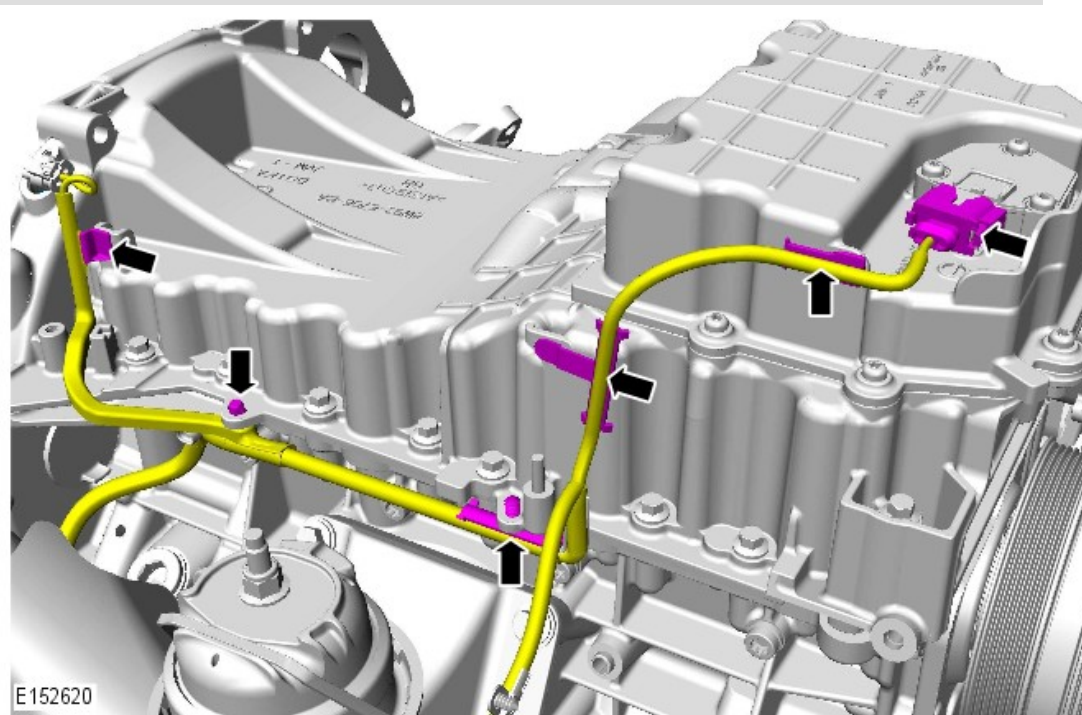


Loosen the Torx screws half a turn each.

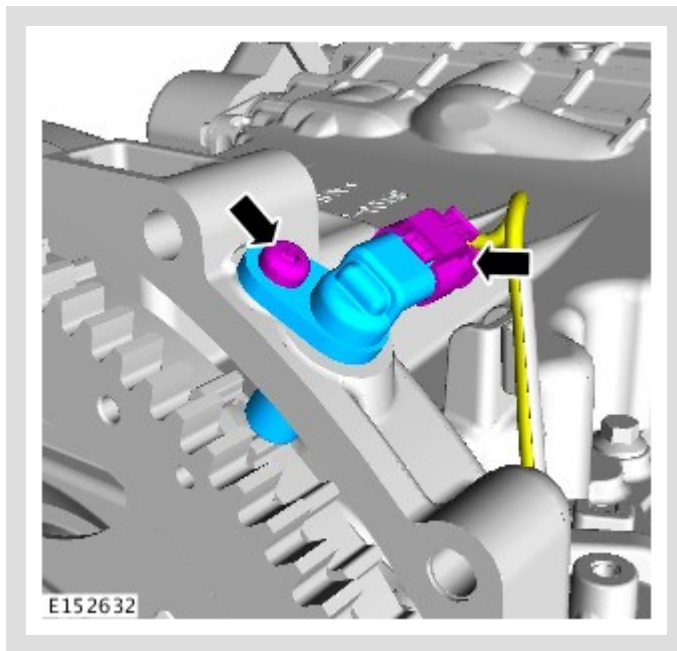
18.



19.



20.



Torque: 10 Nm

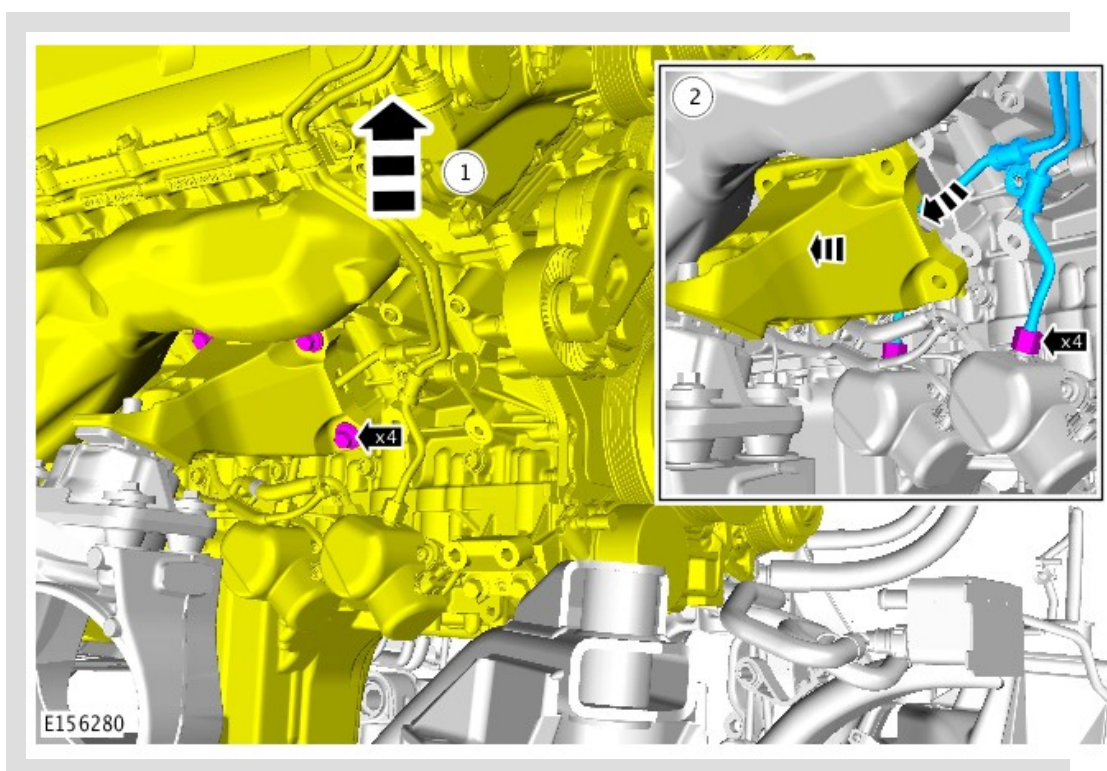
21.

⚠ CAUTION:

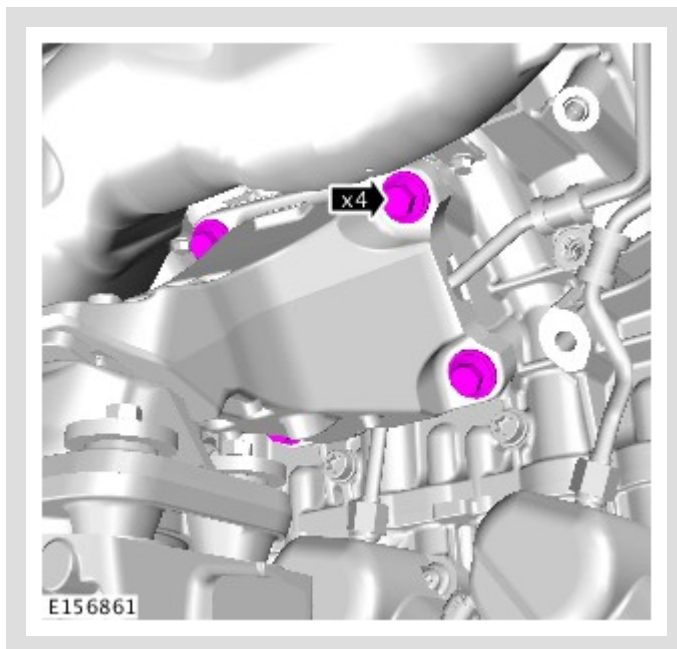
Use a wooden block to protect the oil pan when supporting the engine.

△ NOTE:

Install the unions finger tight before final tightening.



Using a suitable hydraulic jack, raise and support the engine to install the fuel pipe.



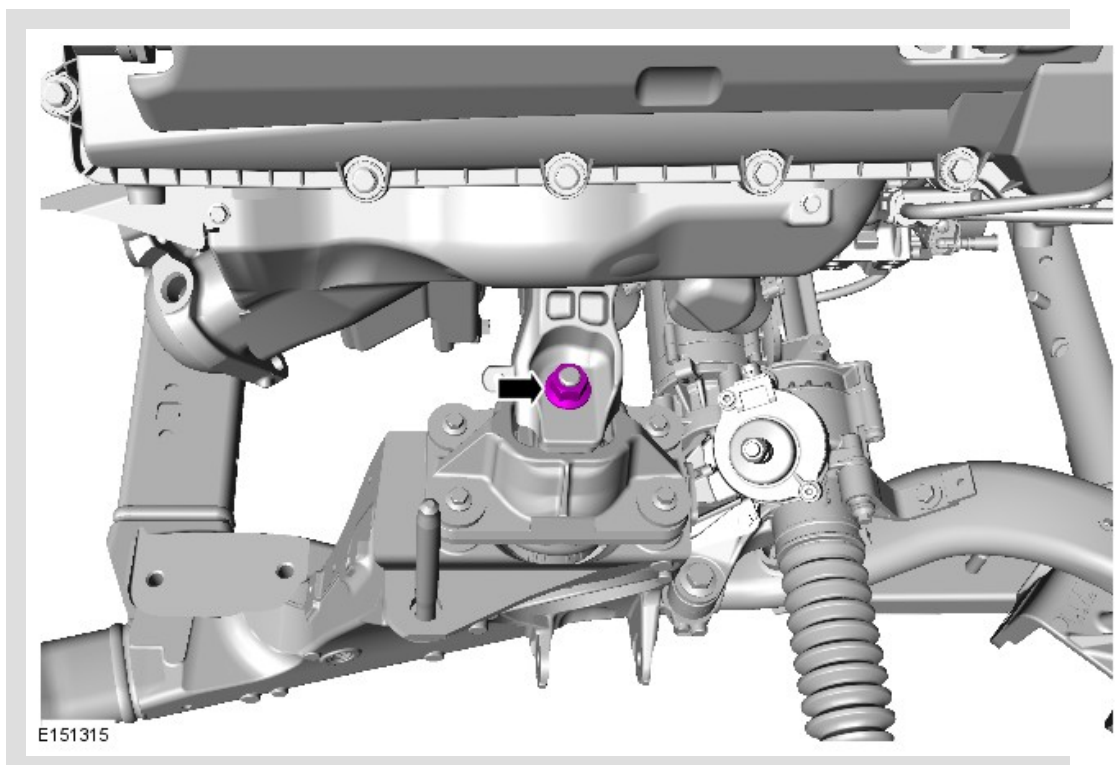
Carefully lower the engine onto the engine mounts.

Torque:

Stage 1: 45 Nm

Stage 2: 60°

23.



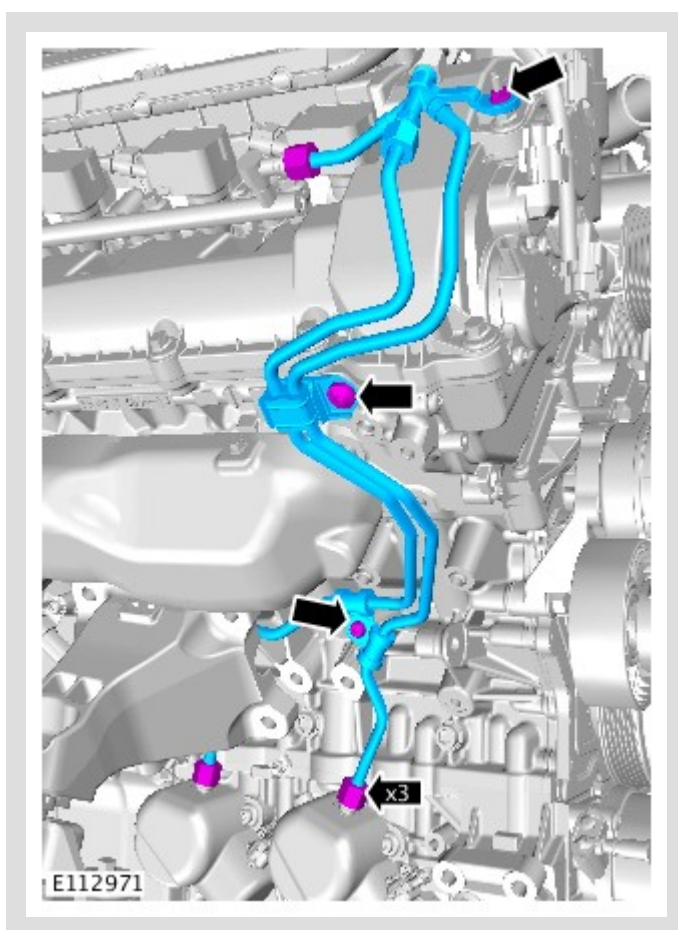
Torque: 110 Nm

**CAUTIONS:**

- Install new high-pressure fuel supply lines.
- Lubricate only the union threads with clean engine oil.

**NOTES:**

- Remove and discard the blanking caps.
- Install the bolt and unions fully finger tight before final tightening.



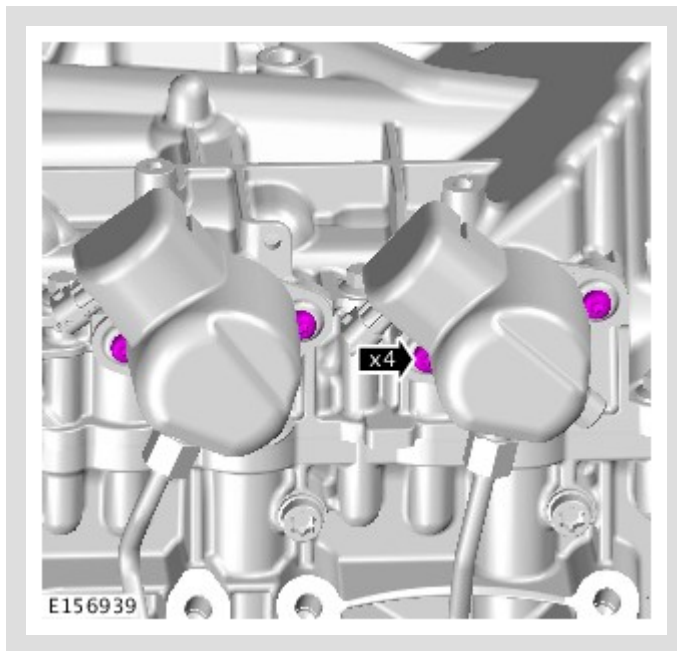
Do not fully tighten at this stage.

Renew Part: *Fuel rail supply pipe(s)* Quantity: *1* .

25.

⚠ CAUTION:

Tighten the Torx screws a turn at a time until the correct torque is achieved.

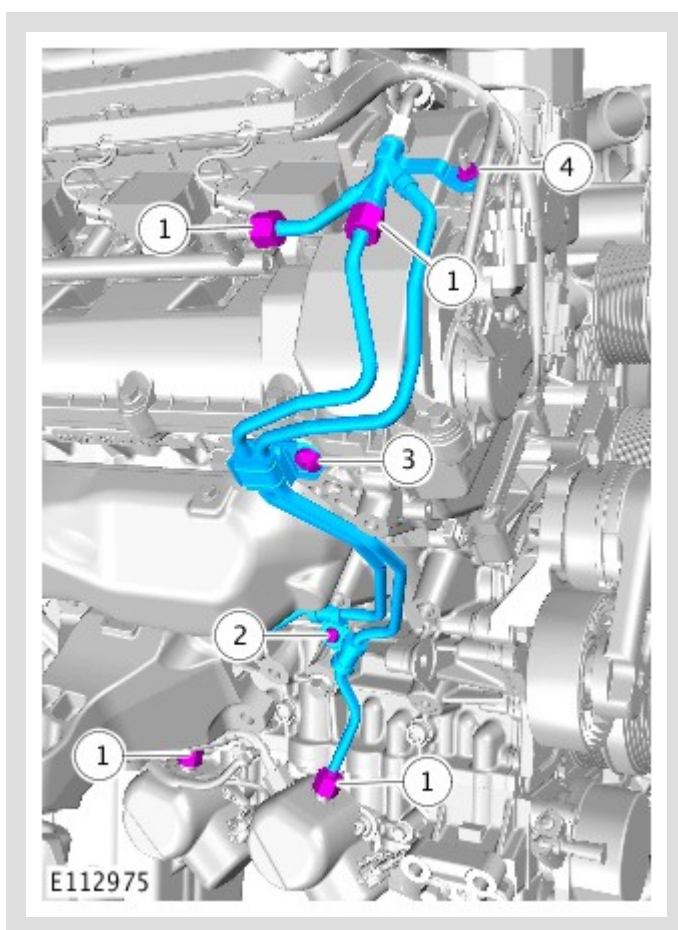


Torque: 12 Nm

26.

 NOTE:

Engine shown removed for clarity.



Torque:

Unions (1) **21 Nm**

M6 (2) **11 Nm**

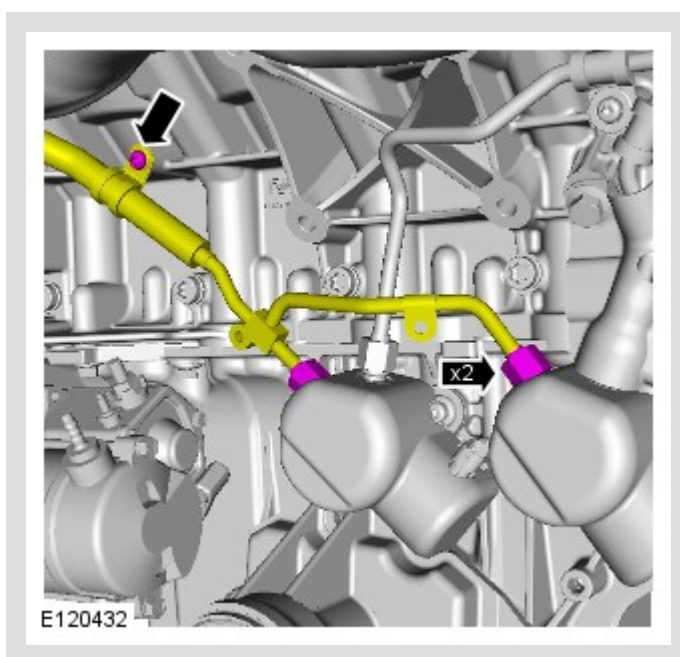
M8 (3) **25 Nm**

M5 nut (4) **6 Nm**

27.

 NOTES:

- Install the bolt and unions finger tight before final tightening.
- Remove and discard the blanking caps.



Torque:

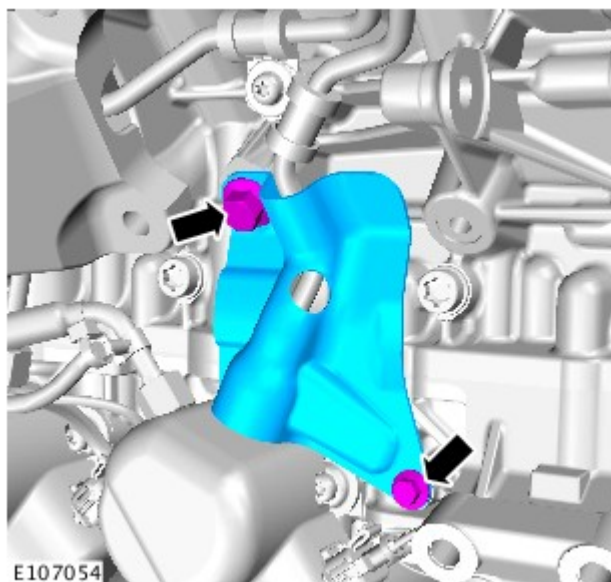
Unions **21 Nm**

M6 **11 Nm**

28.

 NOTE:

Engine shown removed for clarity.



Torque:

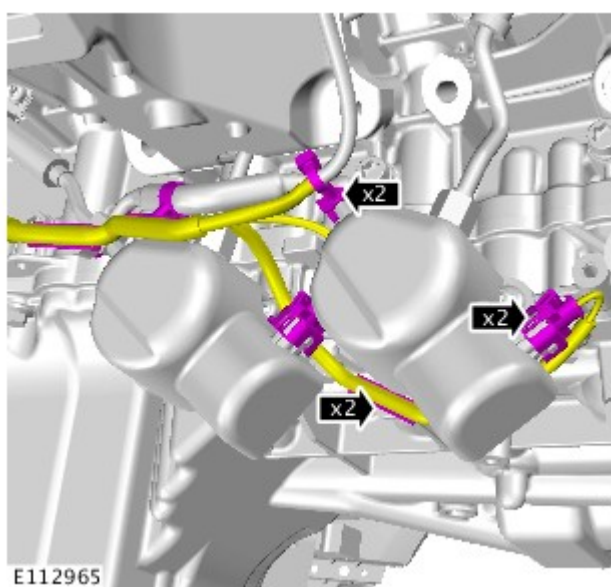
M10 29 Nm

M6 11 Nm

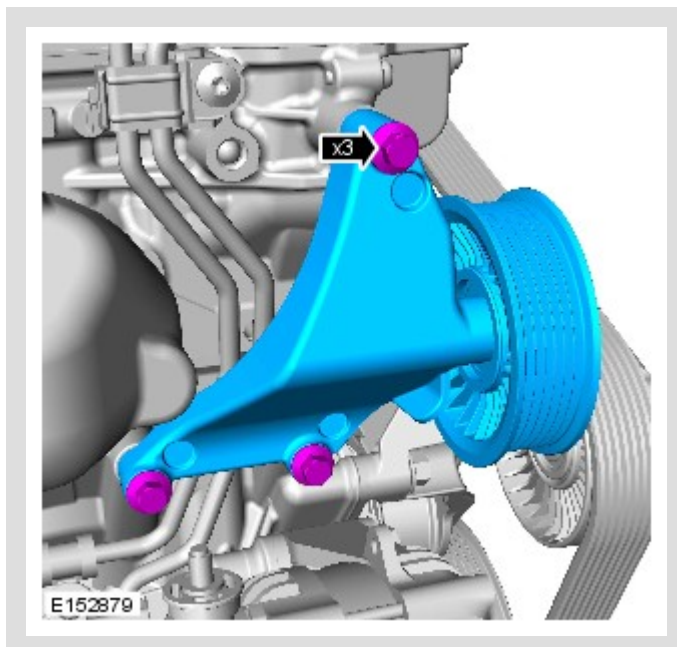
29.

 **NOTE:**

Engine shown removed for clarity.

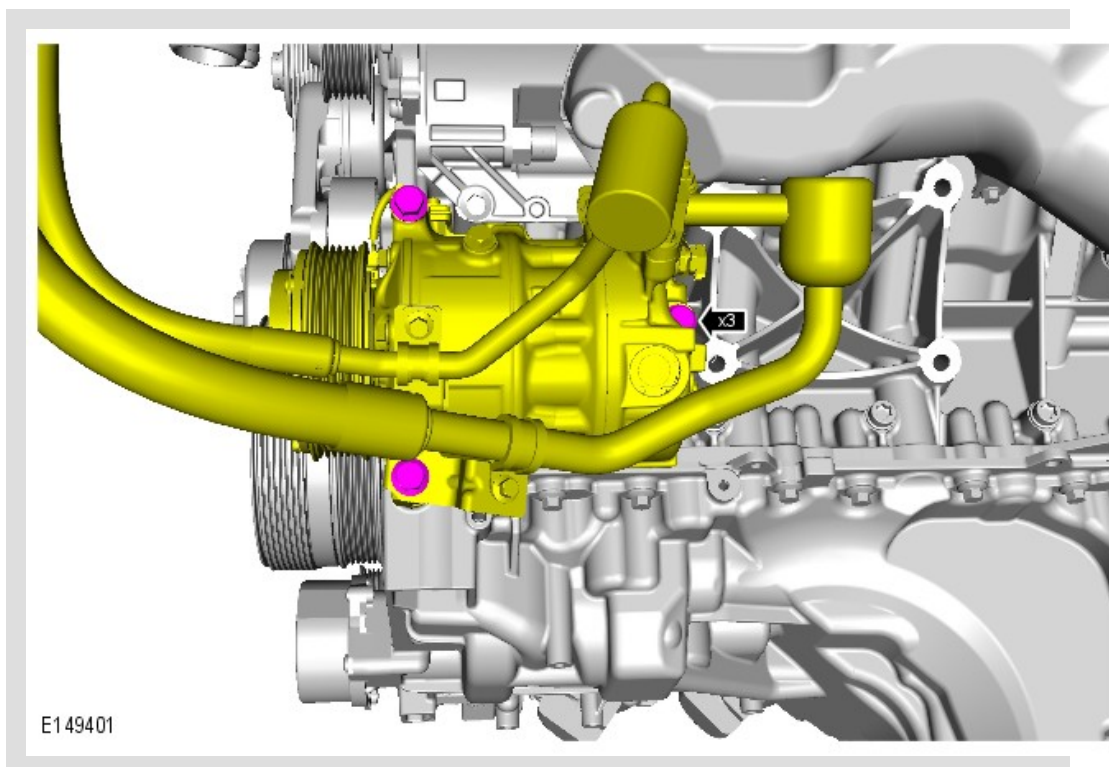


30.



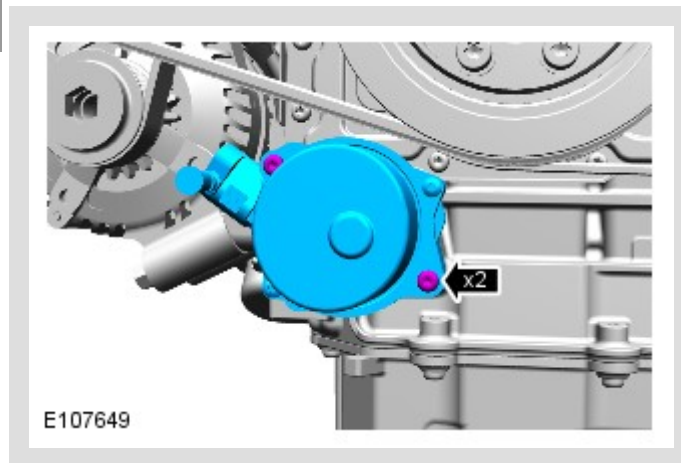
Torque: 25 Nm

31.



Torque: 25 Nm

32.



Torque: 12 Nm

33.



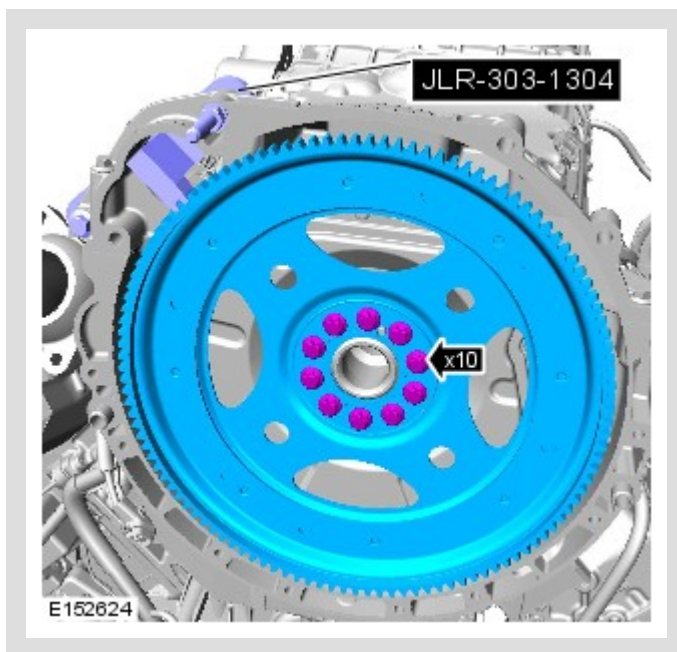
CAUTIONS:

- Install all the bolts finger tight before final tightening.
- The bolts can only be used 3 times, mark the bolts with a centre punch. If 2 punch marks are visible, discard the bolts.



NOTES:

- Make sure the crankshaft and flexplate mating faces are clean before installation.
- Tighten the retaining bolts working diagonally.



Renew Part: *Flexplate bolts* Quantity: *1* .

Torque:

Stage 1: **45 Nm**

Stage 2: **90°**

34. Refer to: Starter Motor (303-06 Starting System - V8 S/C 5.0L Petrol, Removal and Installation).

35. Refer to: Transmission - V6 S/C 3.0L Petrol /V8 S/C 5.0L Petrol, Vehicles With: Twin Speed Transfer Case (307-01 Automatic Transmission/Transaxle - Vehicles With: 8HP70 8-Speed Automatic Transmission AWD, Installation).

36. Refer to: Differential Case (205-03 Front Drive Axle/Differential, Removal and Installation).

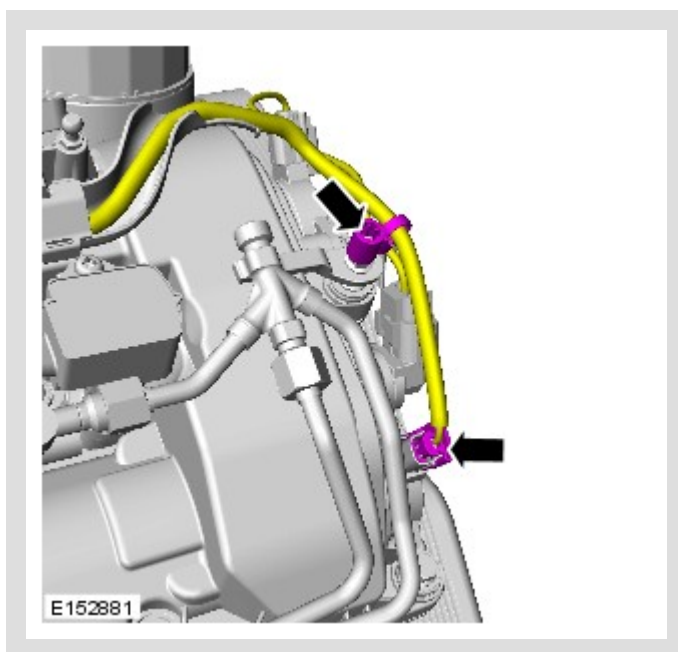
37. Refer to: Generator (414-02 Generator and Regulator - V8 S/C 5.0L Petrol, Removal and Installation).

38.  **WARNING:**

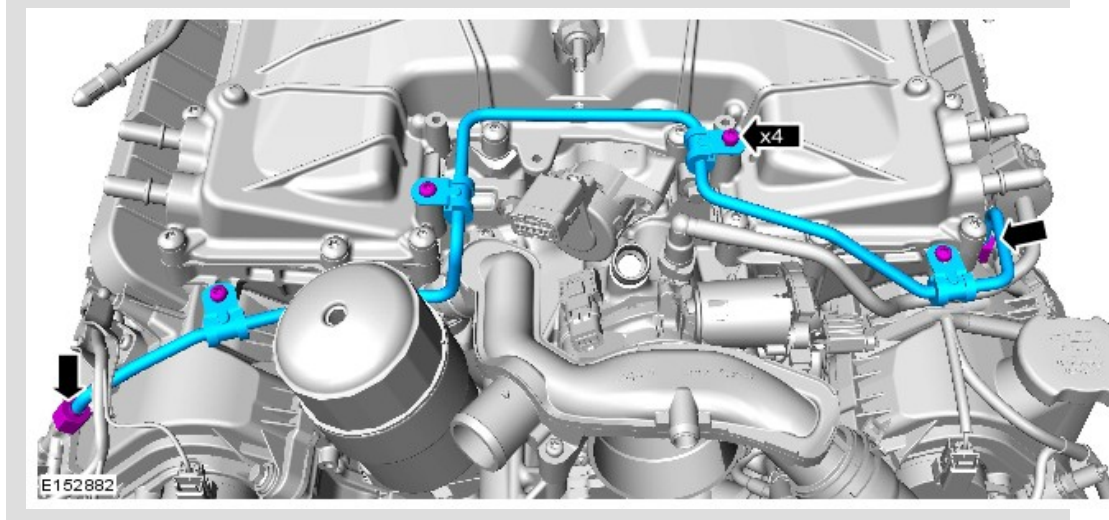
Do not work on or under a vehicle supported only by a jack. Always support the vehicle on safety stands.

Raise and support the vehicle.

39.



40.



Renew Part: *Fuel injector high pressure pipe(s)* Quantity: **1** .

Torque:

Unions **21 Nm**

M6 **8 Nm**

41.

Refer to: Engine Oil Draining and Filling (303-01 Engine - V8 S/C 5.0L Petrol, General Procedures).

42.

Connect the battery ground cable.

Refer to: Specifications (414-01 Battery, Mounting and Cables, Specifications).